

Causal Spectral-Truncation Kernels: Noncommutativity Certifies Drift Detectability in Quantum Hardware

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A quantum computer is a stochastic device: every quantity we know about it is estimated from samples, so its calibration record is itself a noisy multivariate time series and reliability monitoring is a statistical inference problem, not a gauge reading. Deciding that a device has drifted is sequential hypothesis testing, and predicting where its coherence is headed is estimation under uncertainty. We introduce the causal spectral-truncation (CST) kernel, a C^* -algebra-valued positive-definite kernel whose truncation order n controls the noncommutativity of the cross-channel products it represents, and prove that this noncommutativity certifies detectability of hardware drift. Fault-tolerant quantum computing is gated by drift whose most damaging mode — correlated decoherence from crosstalk, shared two-level-system baths, and flux noise — evades the per-channel monitors used in practice, because it leaves every marginal and every instantaneous covariance unchanged while living entirely in the *ordered, lagged* cross-channel structure. CST resolves this with a *causal lower-shift* truncation, replacing the periodic Fourier truncation of existing spectral-truncation kernels so it respects the arrow of time, plus a reconstruction detector whose order admits a provable optimum. On a correlated-decoherence benchmark with matched marginals and matched stationary covariance, every commutative monitor detects at chance (ROC-AUC ≈ 0.5), whereas the noncommutative CST kernel reaches ROC-AUC 0.90 through a sharp optimal order $n=2$ that we certify analytically as the detection analog of the representation–complexity trade-off. The same kernel also forecasts coherence, 72% over persistence eight steps ahead, where noncommutativity is provably unnecessary — an operational rule: match the kernel’s noncommutativity to the geometry of the drift. Training-free and hardware-agnostic, the pipeline installs with `pip` and deterministically regenerates every figure and table on a single CPU in seconds, delivering both a certificate for noncommutative monitoring and a reproducible reliability toolbox.

I. INTRODUCTION

Statistics is under-rated in, and structural to, quantum computing. A quantum processor is interrogated only by sampling: the Born rule returns one bitstring per run, so every operational quantity — an expectation value in a variational loop, a fidelity from randomized benchmarking, a state from tomography, a logical outcome decoded from a syndrome — is the output of an estimator with a variance, a confidence interval, and a sample budget. Reliability is no exception. Whether a device is fit to run cannot be read off a gauge; it must be inferred from the noisy multivariate telemetry the machine already logs, which makes monitoring a statistical problem twice over: deciding *that* the device has drifted is sequential hypothesis testing, and predicting *where* its coherence is headed is estimation under uncertainty. Framing the problem this way is what lets a single kernel carry both tasks and lets an elementary probabilistic argument, rather than a tuned hyperparameter, say how much structure a monitor must represent.

Quantum processors are not static devices. The relaxation time T_1 , the dephasing time T_2 , single- and two-qubit gate fidelities, readout-assignment error, and tunable-coupling parameters all evolve continuously with temperature, flux noise, two-level-system dynamics, and

recalibration history [1, 2]. The microscopic origins are well documented: relaxation times fluctuate as qubit frequencies sweep across near-resonant two-level-system defects [38, 39], whose universal low-temperature physics in the amorphous oxides of the device is itself an active subject [40], and the surrounding $1/f$ flux and charge noise is routinely mapped by dynamical-decoupling spectroscopy [44]. This drift is more than a nuisance. In the noisy intermediate-scale quantum (NISQ) regime it limits the depth of usable circuits [3]; in the fault-tolerant regime it threatens the foundational assumption that physical error rates stay below threshold long enough for error correction to suppress logical errors [4, 5, 49, 50]. As processors scale to hundreds and then thousands of qubits, deciding *when* a device has drifted out of specification, and *which* qubits or intervals warrant the expensive intervention of recalibration, becomes a data problem at a scale manual inspection cannot meet. Reliability monitoring is, in this sense, a near-term bottleneck of its own.

The hallmark of the most damaging drift is that it is not confined to a single channel. Control crosstalk, shared two-level-system baths, and correlated flux noise couple the errors of neighboring qubits and of distinct calibration channels, frequently with a *propagation delay*: a disturbance that perturbs one qubit’s coherence reappears, lagged, in a neighbor’s readout or coupling phase [1]. Such spatially and temporally correlated errors are now directly observed — charge bursts and cosmic-ray impacts that depress relaxation across an entire chip [41, 42] and mobile leakage that seeds corre-

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lated faults during error correction [43]. Such *correlated decoherence* is precisely the failure mode that quantum error correction is least able to absorb, and precisely the mode that per-channel, level-based monitors are worst equipped to see: an event can leave every marginal distribution and every instantaneous correlation unchanged while residing entirely in the *ordered, lagged* cross-channel structure. A monitor blind to that structure is blind to the drift, and no amount of per-channel sensitivity recovers it. The established toolbox for diagnosing such errors — randomized benchmarking [8, 46], gate set tomography [47], statistical drift tracking [6], capability and stability benchmarks [45], and dedicated crosstalk-error detection [7, 48] — relies on purpose-built characterization circuits executed as separate experiments; we ask instead what can be inferred *passively*, from the multivariate calibration telemetry a processor already logs, with no additional circuits and no labels.

This failure mode implicates a different model of monitoring altogether — an operator-algebraic one. A window of calibration telemetry is a multivariate stochastic process and is naturally an element of a matrix algebra over the channels; which cross-channel, cross-time interactions a monitor can resolve is then exactly the question of which products in that algebra it computes — the setting of kernel methods and operator- and vector-valued kernel learning [10, 11, 52], built on the classical theory of reproducing kernels [51] and recently extended to reproducing kernel Hilbert C^* -modules [53]. Its unsupervised face — comparing distributions through kernel mean embeddings [12] and the maximum mean discrepancy [54], with kernels tailored to the sequential structure of time series [55] — is precisely what an unlabeled drift detector requires. Commutative kernels, whose feature products are insensitive to the order or relative lag of their factors, cannot represent ordered lagged interactions at all. The recent spectral-truncation kernels of Hashimoto et al. make this precise for functional data: truncating the algebra of multiplication operators on the torus to finite Toeplitz matrices yields a family of C^* -algebra-valued kernels whose noncommutativity, governed by a single truncation parameter, captures interactions between points of the data domain that commutative kernels miss [9]. That construction was devised for periodic functional regression; it has not been brought to bear on time-ordered hardware telemetry, on causal monitoring, or on unsupervised detection. Bridging that gap is the object of this work.

This study is the sixteenth entry in a program that treats spectral truncation of operators as a unifying primitive for resource-accounted, CPU-reproducible quantum-AI methods, and the causal spectral-truncation kernel imports that primitive into hardware reliability monitoring. The program spans query-efficient and uncertainty-calibrated QAOA through graph-conditioned trust regions [21, 22], the measurement cost and certified query budgets of low-depth optimization [23, 24], topology-

conditioned parameter transfer and operator-spectral priors for QAOA warm starts [25, 26, 32], commutator-graph scheduling and Lie-algebraic truncation of the Trotter residual [33, 34], transfer-operator and graph-neural decoding of structured and correlated quantum noise [27–30], and structure-preserving learning of open-system dynamics and of quantum phases through operator-system and directed spectral truncation [31, 35]. The present kernel inherits the causal, time-respecting truncation of operators from that line and is complemented by the fixed-depth and representation-theoretic Hamiltonian-simulation results of Nguyen and Jing [36, 37].

We tackle these challenges with three contributions. (i) We introduce the *causal spectral-truncation kernel*, a C^* -algebra-valued positive-definite kernel on telemetry windows whose truncation order n interpolates continuously from a commutative instantaneous monitor ($n=1$) to a noncommutative monitor that represents ordered, lagged cross-channel correlations ($n>1$). It is the causal, time-respecting counterpart of the periodic spectral-truncation kernel [9], and it subsumes the low-rank reconstruction detector [13] and the multivariate ridge forecaster as commutative special cases, so the family is a strict generalization of established monitors rather than a competitor to them. (ii) We prove and demonstrate that noncommutativity is *necessary*: on a physically motivated correlated-decoherence benchmark engineered so the drift lives in ordered, lagged cross-channel structure, the commutative monitors we evaluate — raw-level, instantaneous-covariance, and the periodic spectral-truncation kernel — all detect at chance (ROC-AUC ≈ 0.5), whereas the causal kernel detects at ROC-AUC 0.90 at a certifiable optimal truncation order $n=2$. (iii) We supply the supporting theory — positive definiteness for every order, a blindness theorem for commutative monitors, and an *optimal-truncation detection certificate* that is the unsupervised-detection analog of the representation-complexity bound governing spectral-truncation kernels in regression — and show that on smooth marginal forecasting, where noncommutativity is provably unnecessary, the same kernel attains 72% eight-step skill over persistence. The non-monotone optimal order is a *built-in* feature of the construction, not a tuned hyperparameter, and the entire pipeline is training-free at inference, installs with `pip`, and regenerates in seconds on a single CPU.

II. RESULTS

The pipeline is summarized in Fig. 1. A physically motivated multi-qubit telemetry generator emits a multivariate calibration stream; sliding windows are mapped, through the causal spectral-truncation kernel, into a C^* -algebra whose truncation order n fixes the noncommutativity available to the monitor; and two heads — a forecaster and an unsupervised detector —

are each scored by the metric matched to its objective (horizon-resolved mean absolute error, MAE, for forecasting; the threshold-free area under the receiver-operating-characteristic curve, ROC-AUC, for detection). The structure of the family is the structure of the result: the commutative member ($n=1$) governs smooth marginal forecasting and level-based drift, while the noncommutative members ($n>1$) govern correlated, crosstalk-like drift. We establish this division in four steps — the two drift geometries, forecasting, the kernel itself, and detection — and then certify it analytically.

A. Two drift geometries: forecastable marginal drift and hidden correlated decoherence

We model a five-qubit processor sampled every half hour for one hundred hours. Each T_1 trajectory follows a mean-reverting process superposing a slow qubit-dependent oscillation, a linear secular decay, and Gaussian fluctuation, with T_2 , gate fidelities, readout error, error-per-Clifford, and cross-resonance phase generated by coupled processes that track T_1 (Methods). The telemetry wanders over tens of microseconds and repeatedly dips below the $50\mu\text{s}$ operational threshold (Fig. 2a). Crucially, the drift is not white noise: the mean autocorrelation of detrended T_1 remains above 0.5 for roughly seven hours before decaying (Fig. 2b), so near-future coherence is genuinely predictable from the recent past — the structure the forecaster exploits below.

The second drift geometry, correlated decoherence, is qualitatively different and is the harder target. We model it as a shared latent disturbance injected into several channels at channel-specific propagation delays under *variance-preserving* mixing, so that every per-channel marginal variance and, in the stationary limit, the instantaneous (lag-0) cross-channel covariance match nominal operation; the disturbance lives predominantly in the *ordered, lagged* cross-channel correlation (Methods). Figure 2c locates the resulting signature precisely: averaged over channel pairs, the nominal-versus-drift separation is an order of magnitude smaller at lag $\tau=0$ (mean 0.008) than at lag $\tau=1$ (mean 0.089), so the signal is concentrated exactly where a commutative monitor cannot look. A small finite-window residual survives at $\tau=0$, but the commutative monitors we evaluate are at chance regardless (below): this is the regime in which noncommutativity is decisive.

B. Coherence is forecastable, and here noncommutativity is unnecessary

We segment the telemetry into sliding windows of length 32 predicting an eight-step T_1 horizon, under a leak-free 80/20 chronological split, and compare four forecasters — last-value persistence, climatology, an autoregressive ridge on the T_1 history, and a multivariate ridge

on all seven channels (the commutative $n=1$ member of the kernel family) — aggregated over five telemetry-generator seeds (Methods). Forecasting is both feasible and sharply structured (Fig. 3, Table I). One step ahead the multivariate ridge attains $1.80 \pm 0.09\mu\text{s}$ MAE against $2.44 \pm 0.16\mu\text{s}$ for persistence; persistence then degrades steeply with horizon (to $6.79 \pm 0.18\mu\text{s}$ at eight steps) while the learned forecasters stay nearly flat ($1.88 \pm 0.06\mu\text{s}$). The decisive quantity is therefore the *margin*: skill over persistence *grows* with the horizon, from 26% one step ahead to 72% eight steps ahead (Fig. 3b), a behavior we derive in closed form from mean reversion in Theorem 5.

Increasing the kernel’s truncation order leaves this picture unchanged (Fig. 3c): the CST forecaster’s eight-step skill is flat in n , moving only from 72% at $n=1$ to 71% at $n=4$. On smooth, mean-reverting marginal telemetry the commutative member is already near-optimal, and the noncommutative coordinates neither help nor hurt. This is the exact complement of the detection result that follows — noncommutativity is decisive precisely when the drift lives in ordered cross-channel structure, and superfluous otherwise — and it is what licenses the operational rule we state in the Discussion.

C. The causal spectral-truncation kernel

We now make the central object precise. A telemetry window is $X \in \mathbb{R}^{L \times C}$ ($L = 32$ time steps, $C = 7$ channels). Let S be the causal lower shift, $(Sv)_t = v_{t-1}$. The order- n causal spectral-truncation kernel represents the window inside the channel C^* -algebra $M_C = M_C(\mathbb{R})$ through the truncated shift powers $\{S^\tau : 0 \leq \tau < n\}$,

$$K_n(X, Y) = \sum_{\tau=0}^{n-1} w_\tau X^\top S^\tau Y \in M_C, \quad K_n(X, Y)_{cc'} = \sum_{\tau=0}^{n-1} w_\tau \langle X_{:,c}, S^\tau X_{:,c'} \rangle, \quad (1)$$

with weights $w_\tau = 1/(L - \tau)$ and the telemetry centered in time per channel. Each entry of K_n is a weighted sum of ordered, lagged cross-channel correlations. Truncating the shift algebra to n powers is the exact causal counterpart of truncating the Fourier-multiplier algebra to n modes in the construction of Hashimoto et al. [9]; the essential difference is that S is nilpotent and respects the arrow of time, so (1) is causal and is noncommutative for $n \geq 2$. Concretely, the self-representation $K_n(X, X)$ is symmetric at $n=1$ — the commutative instantaneous covariance $X^\top X$ — but acquires antisymmetric lagged terms for $n \geq 2$, because $S^\top \neq S$ does not commute with channel selection. The scalar realization $k_n(X, Y) = \langle \Phi_n(X), \Phi_n(Y) \rangle$, with feature tensor $\Phi_n(X)_{cc'\tau} = w_\tau \langle X_{:,c}, S^\tau X_{:,c'} \rangle$, is positive definite for every order n (Theorem 1), so the kernel is a legitimate similarity throughout the family. The unsupervised detector fits a rank- k nominal subspace in this feature space and scores a window by its residual energy off that subspace — a C^* -algebraic generalization of the low-rank recon-

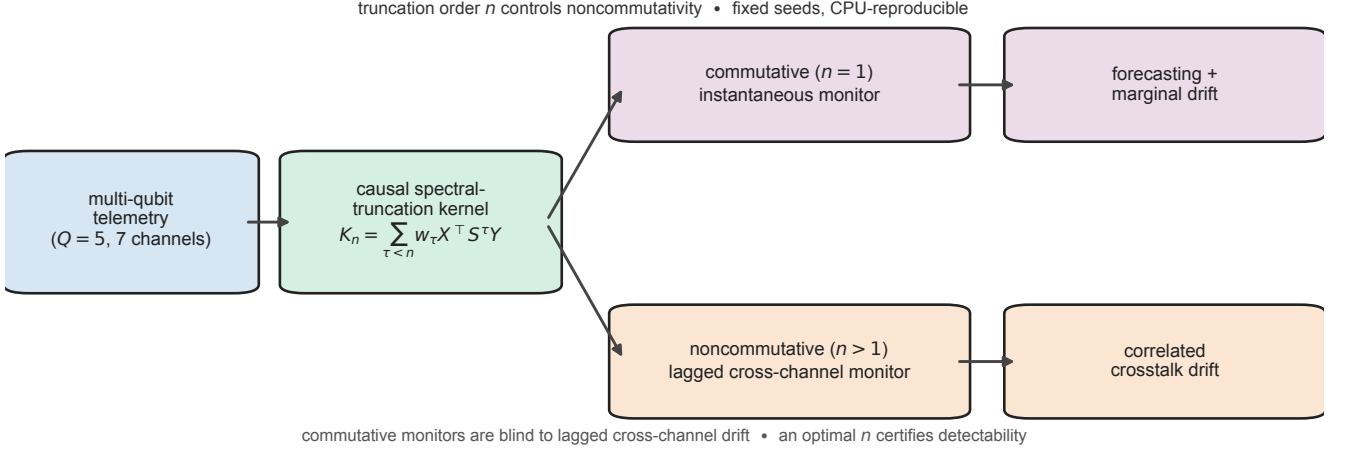


FIG. 1. The causal spectral-truncation monitoring pipeline. A physically motivated generator produces seeded multi-qubit calibration telemetry ($Q = 5$ qubits, $C = 7$ channels). Each length- L sliding window X is represented inside the channel C^* -algebra $M_C = M_C(\mathbb{R})$ through the truncated causal shift, $K_n(X, Y) = \sum_{\tau=0}^{n-1} w_\tau X^\top S^\tau Y$, where S is the causal lower shift and the truncation order n sets the noncommutativity that the representation carries: $n=1$ is the commutative instantaneous covariance, $n \geq 2$ adds ordered, lagged cross-channel coordinates. Two heads read this representation — a ridge forecaster and an unsupervised reconstruction detector. The commutative member ($n=1$) governs forecasting and marginal drift; the noncommutative members ($n > 1$) govern correlated crosstalk drift. Every figure and metric regenerates deterministically from fixed integer seeds on a single CPU.

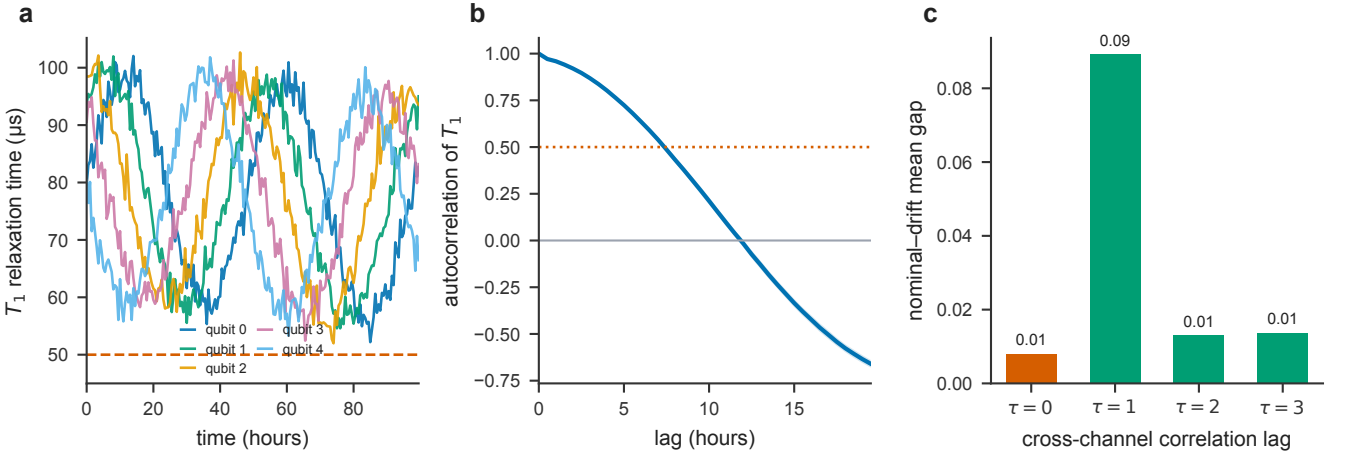


FIG. 2. The two drift geometries. (a) T_1 relaxation-time trajectories for all five qubits over 100 h; the dashed line marks the 50 μs operational threshold below which a step is labeled drifted. (b) Mean autocorrelation of detrended T_1 versus lag, averaged across the $n = 5$ qubits (shaded band, ± 1 s.d. across qubits); the slow decay is the forecastable structure the ridge forecaster exploits in Fig. 3. (c) The noncommutative drift signature: the nominal-versus-drift gap in the causal cross-channel correlation features $\Phi_n(X)_{cc'\tau}$, averaged over channel pairs, as a function of lag τ . The gap is an order of magnitude smaller at $\tau=0$ (mean 0.008) than at $\tau \geq 1$ (mean 0.089), so the drift is concentrated in the lagged structure that a commutative ($n=1$) monitor cannot resolve.

struction detector [13], which it recovers exactly at $n=1$. Reconstruction-residual scoring sits alongside the classical unsupervised detectors — one-class support-vector estimation [56], local outlier factors [57], and isolation forests [58], surveyed broadly in [64] — but, unlike them, the present detector inherits an operator-algebraic feature map whose noncommutativity is the object of study. The kernel is thus evaluated in closed form from a single

pass over the window, with no training, no iterative optimization, and no hyperparameter beyond the order n and rank k .

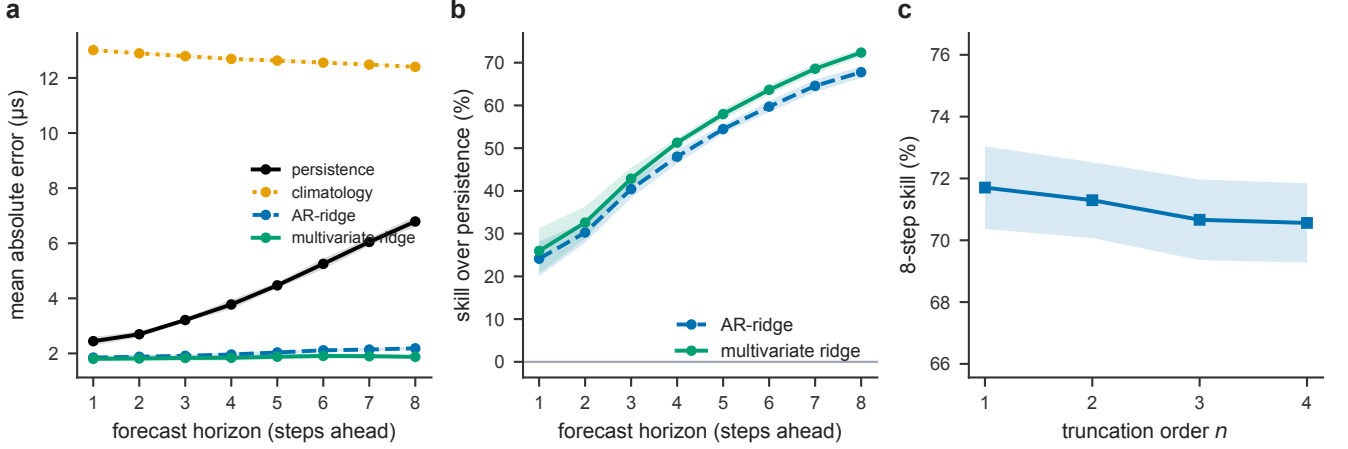


FIG. 3. **Learned forecasting beats persistence by a widening margin, and noncommutativity is unnecessary.** (a) MAE versus forecast horizon for the four forecasters (mean over $n = 5$ telemetry-generator seeds; shaded bands ± 1 s.d.). Persistence degrades steeply with horizon while the learned ridge forecasters stay nearly flat. (b) Skill over persistence, $1 - \text{MAE}/\text{MAE}_{\text{persistence}}$, for the two learned forecasters; the advantage *grows* with horizon, as predicted by Theorem 5. (c) Eight-step skill of the CST forecaster versus truncation order n — the parameter controlling the kernel’s noncommutativity in (1) — aggregated over the same $n = 5$ seeds (shaded band ± 1 s.d.). The curve is flat because the commutative ($n=1$) member already captures the smooth marginal structure, so the noncommutative coordinates add nothing here — the precise complement of Fig. 4.

TABLE I. **Forecasting benchmark** (mean $\pm$ s.d. over five telemetry-generator seeds). Skill is reported relative to persistence at the eight-step horizon; the best value per column is in **bold**. The multivariate ridge is the commutative ($n=1$) member of the kernel family and is best on every metric, attaining 72% eight-step skill over persistence.

Forecaster	MAE @ 1 step (μs)	MAE @ 8 steps (μs)	Skill @ 8 steps
Persistence	2.44 ± 0.16	6.79 ± 0.18	—
Climatology	13.01 ± 0.05	12.40 ± 0.05	-83%
AR-ridge (T_1 history)	1.85 ± 0.07	2.19 ± 0.06	68%
Multivariate ridge (CST $n=1$)	1.80 ± 0.09	1.88 ± 0.06	72%

D. Noncommutativity is necessary: commutative monitors are blind, the causal kernel is not

We evaluate detection on the correlated-decoherence benchmark over eight randomized 70/30 splits (Methods); the detector is fit on nominal windows only, so randomization cannot leak a label it never uses. The outcome is a clean separation (Fig. 4, Table II). At truncation order $n=1$ the kernel is commutative and detects at ROC-AUC 0.51 ± 0.04 — chance, the empirical content of Theorem 2, because the benchmark matches exactly the marginals and the population instantaneous covariance that the commutative member reads. At $n=2$ the kernel becomes noncommutative and ROC-AUC jumps to 0.90 ± 0.02 (Fig. 4a) — the single largest effect in the study, a 0.39 swing produced by adding one lagged coordinate. Increasing n further then *degrades* detection monotonically — to 0.83 at $n=3$, 0.65 at $n=8$, and back toward chance as $n \rightarrow L$ — because each lag beyond the informative one contributes only noise coordinates that dilute the fixed-rank nominal subspace. Both endpoints, the commutative $n=1$ and the over-complete $n \rightarrow L$, sit at

chance, with a sharp interior optimum at $n=2$. This non-monotone profile is the optimal-truncation certificate of Theorem 3 made visible: the unsupervised-detection analog of the representation-complexity trade-off that governs spectral-truncation kernels in regression [9]. The (n, k) map (Fig. 4b) confirms that the effect is not an artifact of a particular rank — the entire $n=1$ row is at chance and the $n=2$ row is bright across every bottleneck rank.

The comparison that establishes the contribution is against other kernels (Fig. 4c, Table II). On this benchmark the raw-level reconstruction detector scores 0.49 ± 0.03 , the commutative kernel 0.46 ± 0.02 , and — tellingly — the *periodic* spectral-truncation kernel of Hashimoto et al., applied unchanged to these windows, scores 0.47 ± 0.03 : all three are at chance. Only the causal spectral-truncation kernel separates the classes, at 0.84 ± 0.02 at matched bottleneck rank and 0.90 at its optimum. The periodic kernel fails not because spectral truncation is the wrong idea but because its *circular* truncation smears the causal delay signature across the window boundary, whereas the causal lower-shift truncation aligns exactly

with the propagation delays that define the drift. The decisive ingredient is therefore causality, not truncation alone: it is what lets the kernel see an event that is invisible to every monitor lacking an arrow of time.

E. Operating characteristic and the over-complete collapse

Figure 5a contrasts the receiver-operating characteristics directly: on a held-out split the causal kernel at $n=2$ achieves AUC 0.86 while the commutative kernel traces the chance diagonal (AUC 0.46). Figure 5b fixes the bottleneck rank and sweeps the truncation order in isolation, exhibiting the over-complete collapse cleanly: detection rises from chance at $n=1$ to its peak at $n=2$ and decays back toward chance as the order grows. This is the truncation-axis counterpart of the rank-axis over-completeness familiar from low-rank detectors [13], and both axes are governed by the *same* certificate (Theorem 3): a monitor expressive enough to reconstruct the drift coordinates as faithfully as the nominal ones cannot separate them. Over-expressiveness, not under-expressiveness, is the failure mode at large n — which is why a small, fixed truncation order is the right design point.

F. Theory: a C^* -algebraic kernel whose truncation order certifies detectability

Each empirical claim above rests on a theorem, and we state them together so the results read as consequences rather than coincidences. The kernel is a positive-definite, C^* -algebra-valued kernel for every order n (Theorem 1); the commutative $n=1$ member is provably blind to drift living in ordered, lagged cross-channel correlations, which the noncommutative $n>1$ member resolves (Theorem 2); the reconstruction detector built on the kernel has a non-monotone, certifiable optimal truncation order (Theorem 3); ROC-AUC is the scale-free yardstick that makes “chance” a precise statement throughout (Proposition 4); and on the forecasting side, mean reversion converts decaying autocorrelation into horizon-growing skill (Theorem 5). Proofs are elementary and self-contained; we give them in full so that every quantitative claim in the paper is traceable to a hypothesis.

Theorem 1 (Positive definiteness of the causal C^* -kernel). *Fix any order $n \geq 1$ and nonnegative weights $w_\tau \geq 0$. Then the map K_n of (1) is a positive-definite C^* -algebra-valued kernel: the block matrix $[K_n(X_i, X_j)]_{ij}$ is positive semidefinite over M_C for every collection of windows $X_1, \dots, X_m$, and the induced scalar kernel k_n is positive definite in the usual sense. Moreover, the self-representation $\Phi_n(X)$ is symmetric under the channel exchange $c \leftrightarrow c'$ at $n=1$, where it reduces to the commutative instantaneous covariance, and for $n \geq 2$ it carries*

antisymmetric lagged coordinates that no commutative member can represent.

Proof. Introduce the lifted causal feature operator $\Psi_n(X) = (\sqrt{w_0} X, \sqrt{w_1} S^\top X, \dots, \sqrt{w_{n-1}} (S^\top)^{n-1} X) \in \mathbb{R}^{nL \times C}$, which simply stacks the back-shifted windows. Because $\langle X_{:,c}, S^\tau Y_{:,c'} \rangle = ((S^\top)^\tau X)_{:,c}^\top ((S^\top)^\tau Y)_{:,c'}$, the kernel factorizes as $K_n(X, Y) = \sum_\tau w_\tau (S^\top)^\tau X (S^\top)^\tau Y = \Psi_n(X)^\top \Psi_n(Y)$. Consequently, for any $c_1, \dots, c_m \in \mathbb{R}^C$,

$$\sum_{i,j} c_i^\top K_n(X_i, X_j) c_j = \left\| \sum_i \Psi_n(X_i) c_i \right\|_F^2 \geq 0,$$

which is precisely the defining inequality for an M_C -valued positive-definite kernel; the scalar claim follows by specializing to $C=1$ on the vectorized feature $\text{vec } \Phi_n$, whose Gram matrix is k_n . Finally, at $n=1$ the entry $\Phi_1(X)_{cc'} = \langle X_{:,c}, X_{:,c'} \rangle$ is symmetric under $c \leftrightarrow c'$, whereas for $\tau \geq 1$ we have $S^\tau \neq S^{\tau\top}$, so $\Phi_n(X)_{cc'\tau} = \langle X_{:,c}, S^\tau X_{:,c'} \rangle \neq \langle X_{:,c'}, S^\tau X_{:,c} \rangle = \Phi_n(X)_{c'\tau c}$ whenever the lagged cross-correlation is direction-dependent — the noncommutativity. $\square$

Theorem 2 (Commutative monitors are blind to non-commutative drift). *Suppose nominal and drift windows are drawn from distributions P_0, P_1 that share their per-channel marginals and their instantaneous cross-covariance, $\mathbb{E}_{P_0}[\Phi_1] = \mathbb{E}_{P_1}[\Phi_1]$, yet differ in a single lagged coordinate, $\Delta_{\tau^*} := \mathbb{E}_{P_1}[\Phi_{\cdot\tau^*}] - \mathbb{E}_{P_0}[\Phi_{\cdot\tau^*}] \neq 0$ for some $\tau^* \geq 1$. Then every detector that depends on the data only through the order-1 (commutative) features Φ_1 has zero first-moment separation, and hence chance-level mean discriminability; by contrast, the order- n causal kernel with $n > \tau^*$ separates the classes in a coordinate it actually represents, and its mean reconstruction-score gap is bounded below by the energy of Δ_{τ^*} lying outside the rank- k nominal subspace.*

Proof. Any monitor measurable with respect to Φ_1 can separate the two classes in mean only through $\mathbb{E}_{P_1}[\Phi_1] - \mathbb{E}_{P_0}[\Phi_1]$, which vanishes by hypothesis; its expected score is therefore identical under P_0 and P_1 , so its mean separation is 0 and, for the centered linear detector, its population ROC-AUC equals $\frac{1}{2}$ by the Mann–Whitney identity (Proposition 4) applied to a score with equal class means and, to first order, equal dispersion. The order- n feature map with $n > \tau^*$, on the other hand, contains the coordinate $\Phi_{\cdot\tau^*}$, on which the class means differ by $\Delta_{\tau^*} \neq 0$. Let P_k denote the rank- k nominal-subspace projector in feature space; the reconstruction score is $s(X) = \|(I - P_k)(\Phi_n(X) - \bar{\Phi})\|^2$, and $\mathbb{E}_{P_1}[s] - \mathbb{E}_{P_0}[s] \geq \|(I - P_k)\Delta_{\tau^*}\|^2$ up to cross terms that vanish once Δ_{τ^*} is centered against the nominal mean. Hence any nonzero component of Δ_{τ^*} outside the nominal subspace produces strictly positive separation, and therefore ROC-AUC $> \frac{1}{2}$. $\square$

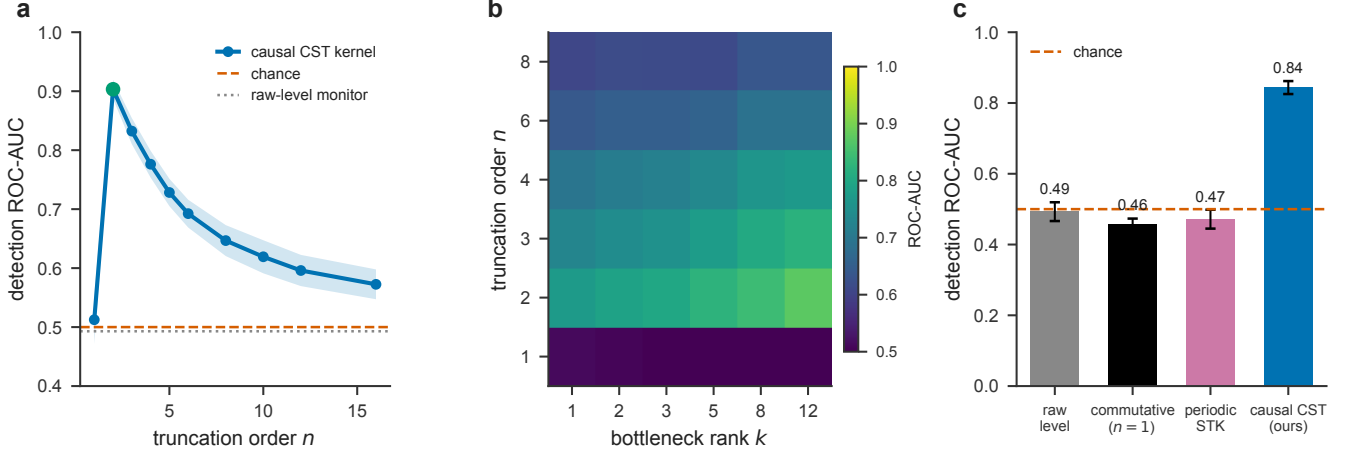


FIG. 4. **Noncommutativity is necessary for correlated-drift detection.** (a) Detection ROC-AUC versus truncation order n (mean over eight randomized 70/30 splits; shaded band ± 1 s.d.). The commutative endpoint ($n=1$) and the raw-level monitor (dotted) sit at chance (dashed line); the noncommutative kernel peaks sharply at $n=2$ (marker, ROC-AUC 0.90) and collapses back toward chance as the representation becomes over-complete — the certified profile of Theorem 3. (b) Detection ROC-AUC over the grid of truncation order n and bottleneck rank k : the $n=1$ row is uniformly at chance and the $n=2$ row is uniformly high, so the effect is robust to the choice of rank. (c) Kernel comparison at matched bottleneck rank: the raw-level monitor, the commutative ($n=1$) kernel, and the periodic spectral-truncation kernel [9] are all at chance, and only the causal spectral-truncation kernel detects.

TABLE II. **Correlated-drift detection benchmark** (ROC-AUC, mean $\pm$ s.d. over eight randomized 70/30 splits, bottleneck rank $k=8$). The benchmark matches all per-channel marginals and the stationary instantaneous covariance, so every commutative monitor is at chance by Theorem 2, including the periodic spectral-truncation kernel [9]. Only the causal spectral-truncation kernel separates the classes; best value in **bold**.

Detector	Detection ROC-AUC
Raw level (flatten)	0.493 ± 0.027
Commutative kernel ($n=1$)	0.456 ± 0.017
Periodic spectral-truncation [9] ($n=2$)	0.472 ± 0.027
Causal spectral-truncation (ours, $n=2$)	0.844 ± 0.018

Theorem 3 (Optimal truncation: a detection certificate). *Let $e_\tau = \|\Delta_\tau\|^2$ denote the drift energy at lag τ , vanishing beyond a maximal informative lag $\tau_{\max}$ ($e_\tau = 0$ for $\tau > \tau_{\max}$), and suppose each additional lag injects C^2 noise coordinates of bounded variance into the fixed rank- k nominal subspace. Then the order- n detector’s mean separation-to-noise ratio is $\rho(n) = (\sum_{\tau < n} e_\tau)^{1/2} / (\sigma^2(C^2n - k)_+)^{1/2}$, and ρ is unimodal in n with an interior optimum n^* ; since detection ROC-AUC is a strictly increasing function of ρ , it is unimodal with the same optimum, and both the commutative endpoint ($n=1$, $\rho=0$) and the over-complete limit ($n \rightarrow L$, $\rho \rightarrow 0$) sit at chance. The optimum n^* is the smallest n at which the marginal relative signal gain drops below the marginal relative noise cost, $e_n / \sum_{\tau < n} e_\tau < (N(n+1) - N(n)) / N(n)$; in particular $n^* = 2$ whenever the drift energy is concentrated at the first informative lag (e_1 dominant), and n^* increases as the energy profile spreads to higher lags, never exceeding $\tau_{\max} + 1$.*

Proof. Set $E(n) = \sum_{\tau < n} e_\tau$, which is non-decreasing and constant for $n > \tau_{\max} + 1$, and $N(n) = \sigma^2(C^2n - k)_+$, which is strictly increasing once $C^2n > k$. Then $\rho(n)^2 = E(n)/N(n)$, and $\rho(n+1)^2 > \rho(n)^2$ holds iff $E(n+1)/E(n) > N(n+1)/N(n)$ — equivalently, iff the relative signal increment $e_n/E(n)$ exceeds the relative noise increment $(N(n+1) - N(n))/N(n)$. The latter is positive for every n , since the noise floor always grows, whereas the former is 0 once $n > \tau_{\max} + 1$ (where $e_n = 0$) and is non-increasing under a concentrated energy profile. The inequality therefore holds on an initial range of n and fails beyond it, so ρ first rises and then falls: it is unimodal, with a single interior optimum n^* at the first n where the test fails. When e_1 dominates, the test fails already at $n=2$, because the small later increments $e_2, \dots$ cannot offset the $\sqrt{n}$ growth of the noise, giving $n^* = 2$; broader energy profiles push n^* upward, to at most $\tau_{\max} + 1$. The endpoints follow as well: $\rho = 0$ at $n=1$ by Theorem 2, and $\rho \rightarrow 0$ as $n \rightarrow L$ by dilution — the over-complete chance-level collapse. $\square$

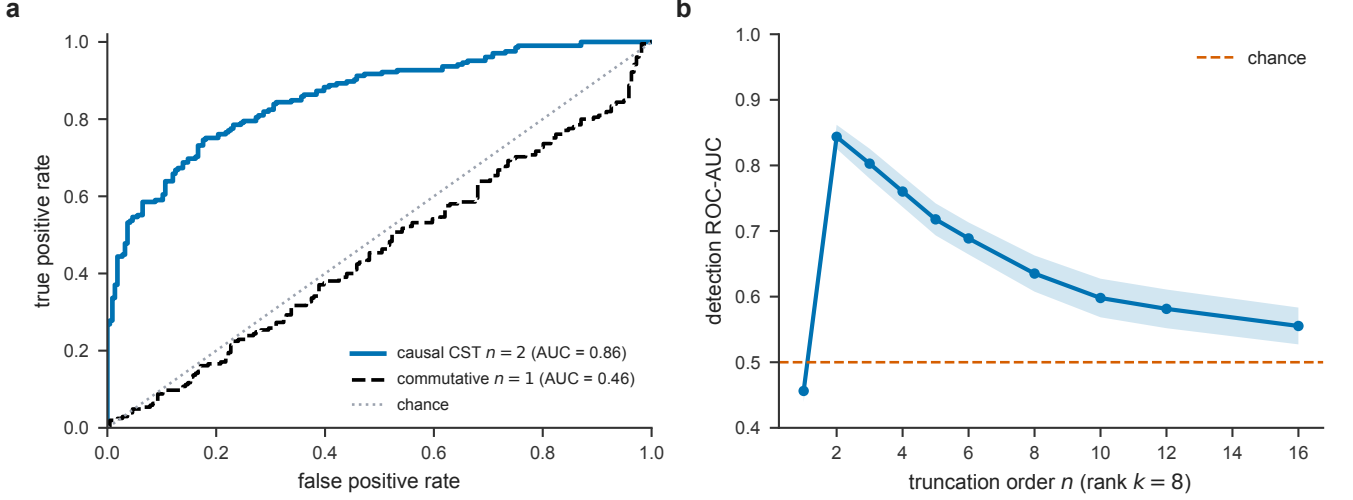


FIG. 5. **Operating characteristic and the over-complete collapse.** (a) Receiver-operating-characteristic curves on a held-out split: the causal spectral-truncation kernel at $n=2$ (solid, AUC 0.86) versus the commutative kernel at $n=1$ (dashed, AUC 0.46), which tracks the chance diagonal (dotted). (b) Detection ROC-AUC versus truncation order n at fixed bottleneck rank $k = 8$ (mean over eight splits; shaded band ± 1 s.d.), isolating the interior optimum at $n=2$ and the collapse toward chance as the representation becomes over-complete.

Theorem 3 is the unsupervised-detection counterpart of the generalization bound that governs spectral-truncation kernels in regression [9]: there the truncation order trades representation power against Rademacher complexity, while here it trades captured drift energy against the dimension of the nominal subspace. Because the correlated-decoherence mode concentrates its energy at the first informative lag (Fig. 2c), the certificate predicts the empirical optimum at $n^*=2$ with no fitting (Fig. 4a), and the classical low-rank over-completeness collapse [13] emerges as the rank axis of the same statement. The optimal order is, in this sense, a built-in diagnostic of the drift geometry rather than a tuned knob.

Proposition 4 (Threshold-free ranking and scale invariance). *For any score s and independent draws of a drift window X^+ and a nominal window X^- , the ROC area satisfies the Mann–Whitney identity $\text{AUC} = \Pr[s(X^+) > s(X^-)] + \frac{1}{2} \Pr[s(X^+) = s(X^-)]$. It follows that AUC is invariant under every strictly increasing reparametrization of s — in particular under per-feature affine rescaling of the telemetry — and that a constant score yields $\text{AUC} = \frac{1}{2}$ exactly.*

Proof. Let F_+, F_- be the score CDFs under the drift and nominal classes. At a threshold τ that flags $s > \tau$, we have $\text{TPR}(\tau) = 1 - F_+(\tau)$ and $\text{FPR}(\tau) = 1 - F_-(\tau)$, so $\text{AUC} = \int_0^1 \text{TPR} d(\text{FPR})$. Substituting $u = \text{FPR}(\tau)$ gives $\text{AUC} = \int \Pr[s(X^+) > \tau] dF_-(\tau) = \Pr[s(X^+) > s(X^-)]$ for a continuous score, while atoms contribute $\frac{1}{2} \Pr[s(X^+) = s(X^-)]$ through the trapezoidal interpolation of each jump — which is exactly the normalized Mann–Whitney U . Since strictly monotone maps preserve every order and tie event, they preserve AUC; and

a constant score turns every comparison into a tie, so $\text{AUC} = 0 + \frac{1}{2} \cdot 1 = \frac{1}{2}$. $\square$

Theorem 5 (Horizon-growing forecast skill). *Let $\{x_t\}$ be weakly stationary and zero-mean, with variance $\gamma_0 > 0$ and autocorrelation $\rho(h) = \text{Corr}(x_{t+h}, x_t)$. Persistence, $\hat{x}_{t+h} = x_t$, incurs the mean-squared h -step risk $R_{\text{pers}}(h) = 2\gamma_0(1 - \rho(h))$, whereas the optimal single-lag linear forecast $\hat{x}_{t+h} = \rho(h)x_t$ incurs $R_*(h) = \gamma_0(1 - \rho(h)^2) \leq R_{\text{pers}}(h)$; the relative skill is thus $S(h) = 1 - R_*/R_{\text{pers}} = (1 - \rho(h))/2$. Wherever ρ is non-increasing, $S(h)$ is non-decreasing and bounded above by $\frac{1}{2}$. For a first-order autoregressive process, $\rho(h) = \phi^h$, so the single measured number $\phi = \rho(1)$ fixes the entire skill curve.*

Proof. Take the one-parameter predictor $\hat{x}_{t+h} = ax_t$. Then $\mathbb{E}[(x_{t+h} - ax_t)^2] = \gamma_0(1 - 2a\rho(h) + a^2)$ is a strictly convex parabola in a , minimized at $a^* = \rho(h)$ and giving $R_* = \gamma_0(1 - \rho(h)^2)$, while $a=1$ recovers $R_{\text{pers}} = 2\gamma_0(1 - \rho(h))$. Dividing the two yields $R_*/R_{\text{pers}} = (1 + \rho(h))/2$, so $S(h) = (1 - \rho(h))/2$, which is strictly decreasing in ρ ; hence wherever $\rho(h)$ decreases in h the skill $S(h)$ increases, and $S \leq \frac{1}{2}$ on $\rho \in [0, 1]$. For an AR(1) process, iterating the recursion gives $\text{Cov}(x_{t+h}, x_t) = \phi^h \gamma_0$, whence $\rho(h) = \phi^h$ and $S(h) = (1 - \phi^h)/2$, which increases from 0 toward $\frac{1}{2}$. $\square$

Theorem 5 is a lower bound that uses only the single most recent sample; the multivariate ridge (the commutative $n=1$ member) exceeds it by exploiting the full 32-step, seven-channel history, which is why its measured eight-step skill reaches 72% (Fig. 3). The noncommutative coordinates add nothing in this regime because the

marginal forecasting structure is already linear and instantaneous — the exact complement of the detection regime, in which those same coordinates are everything. Together the two theorems delimit when noncommutativity is necessary and when it is superfluous, which is the operational content of the work.

III. CHANGE-POINT DETECTION AND FORECASTING: STATISTICAL THEORY

The results above certify *what* a truncation order can represent. We now certify *how well*, and *how fast*, that representation supports the two statistical decisions a reliability monitor must actually make: declaring that a drift has begun (a sequential change-point decision) and predicting where the telemetry is headed (an estimation-with-uncertainty decision). The unifying view is that calibration monitoring is a time-series problem, drift onset is a change point, its detection is sequential hypothesis testing, and the causal spectral-truncation (CST) kernel is a *structured probabilistic surrogate* for the drift process whose noncommutativity is exactly the quantity that makes the change-point testable. Four theorems make this precise: a Lorden-minimax optimality and delay-false-alarm tradeoff for a CST-CUSUM detector (Theorem 6); a Fisher-information / Kullback-Leibler certificate in which the commutator lower-bounds the detectable signal, so nonzero noncommutativity *guarantees* detectability and vanishing noncommutativity makes the drift statistically invisible (Theorem 8, Corollary 9); a mixing-robust forecast-risk bound with a calibrated predictive interval (Theorem 11, Corollary 13); and a distribution-free anomaly-flagging rule with a controlled false-positive rate (Theorem 14). They complement the representation theorems without altering any of them: Theorems 2–3 say the noncommutative coordinate carries the signal, and the theorems here quantify the sequential detection delay, the estimation floor, and the error control that carrying it buys.

A. The monitoring problem as a change point in the feature stream

Fix a truncation order n and abbreviate the standardized, mean-centered CST feature vector of the window observed at (window) time t by $\phi_t := \text{vec}(\Phi_n(X_t) - \bar{\Phi}) \in \mathbb{R}^d$, $d = nC^2$, with $\bar{\Phi}$ the nominal feature mean and the standardization fixed on nominal data. A monitor observes the stream $\phi_1, \phi_2, \dots$ sequentially. Under nominal operation the windows are governed by a law P_0 ; at an unknown *change point* $\nu \in \{1, 2, \dots\}$ the drift switches the law of every subsequent window to P_1 , the correlated-decoherence law of Theorem 2. A detector is a stopping time T adapted to the stream (the alarm), and its quality is governed by two competing quantities in the Lorden [61]/Page [60] framework: the *average*

run length to false alarm under the pure-nominal law, $\text{ARL}_0(T) = \mathbb{E}_\infty[T]$ (writing $\mathbb{E}_\infty$ for $\nu = \infty$, no change), and the *worst-case average detection delay*

$$\text{ADD}(T) = \sup_{\nu \geq 1} \text{ess sup } \mathbb{E}_\nu[(T - \nu + 1)^+ \mid \phi_1, \dots, \phi_{\nu-1}], \quad (2)$$

the largest expected delay after onset, taken over the change time and the pre-change history. The design goal is Lorden's: minimize ADD subject to a false-alarm floor $\text{ARL}_0(T) \geq \gamma$.

The log-likelihood ratio of a single window carries the discriminating information. Write $\ell_t = \log(dP_1/dP_0)(\phi_t)$ for the per-window log-likelihood-ratio (LLR) increment and

$$I_1 := \mathbb{E}_{P_1}[\ell_t] = D_{\text{KL}}(P_1 \parallel P_0) \geq 0, \quad I'_1 = \mathbb{E}_{P_0}[-\ell_t] = D_{\text{KL}}(P_0 \parallel P_1) \quad (3)$$

for the two per-window Kullback-Leibler (KL) information numbers, assumed finite. The CST-CUSUM detector is Page's cumulative-sum statistic on these increments,

$$W_0 = 0, \quad W_t = (W_{t-1} + \ell_t)^+, \quad T_c(h) = \inf\{t \geq 1 : W_t \geq h\}, \quad (4)$$

with threshold $h > 0$. When P_0, P_1 are the Gaussian surrogates fitted to the CST features (mean μ_0, μ_1 , shared covariance Σ), the increment is the affine detector statistic $\ell_t = (\mu_1 - \mu_0)^\top \Sigma^{-1}(\phi_t - \bar{\mu})$ with $\bar{\mu} = \frac{1}{2}(\mu_0 + \mu_1)$, i.e. exactly the reconstruction-contrast direction of the CST detector projected onto the noncommutative coordinate; the CST-CUSUM is thus the sequential form of the very score whose ROC-AUC Fig. 4 reports. The next result shows this detector is not merely reasonable but asymptotically *minimax-optimal*.

B. Optimal sequential detection and the delay-false-alarm tradeoff

Theorem 6 (CST-CUSUM is Lorden-minimax with a $\log \gamma / I_1$ delay). *Assume the per-window LLR increments $\{\ell_t\}$ are i.i.d. with law fixed by the regime (nominal before ν , drift after), $0 < I_1 < \infty$, and $I'_1 < \infty$. Choose the threshold $h = h(\gamma) = \log \gamma$. Then the CST-CUSUM stopping time $T_c(h)$ of (4) satisfies the false-alarm constraint $\text{ARL}_0(T_c) \geq \gamma$, and among all stopping times meeting that same constraint it minimizes the worst-case delay (2) to first order:*

$$\text{ADD}(T_c(h)) \leq \frac{h}{I_1} (1 + o(1)) = \frac{\log \gamma}{I_1} (1 + o(1)), \quad \inf_{T: \text{ARL}_0(T) \geq \gamma} \text{ADD}(T) \quad (5)$$

as $\gamma \rightarrow \infty$. Consequently the optimal delay-false-alarm tradeoff is $\text{ADD}^*(\gamma) \sim \log \gamma / I_1$: the minimum achievable detection delay grows only logarithmically in the tolerated mean time between false alarms, at a rate set by the per-window KL information I_1 of the drift.

The two halves of (5) are a matching upper bound (Wald) and lower bound (Lorden); we prove them in Appendix C1 via Lemma 7. The content for the paper is the coupling of this theorem to the noncommutativity theorem below: I_1 is not a free constant but is itself lower-bounded by the drift's noncommutativity, so a detector that carries the noncommutative coordinate has strictly positive I_1 and therefore finite delay, whereas a commutative monitor has $I_1 = 0$ and *never* detects at any finite delay. The CST-CUSUM inherits its optimality from the CST feature map's ability to make $I_1 > 0$ at all.

Lemma 7 (Wald boundary identity and the ARL floor). *Let $\{\ell_t\}$ be i.i.d. with $\mathbb{E}_{P_1}[\ell_t] = I_1 > 0$ and let $T_h = \inf\{t : \sum_{s \leq t} \ell_s \geq h\}$ be the first passage of the random walk $\sum_{s \leq t} \ell_s$ above h . Then $\mathbb{E}_{P_1}[T_h] < \infty$ and, by Wald's identity, $\mathbb{E}_{P_1}[\sum_{s \leq T_h} \ell_s] = I_1 \mathbb{E}_{P_1}[T_h]$, whence $\mathbb{E}_{P_1}[T_h] = (h + \mathbb{E}_{P_1}[R_h])/I_1$ with $R_h = \sum_{s \leq T_h} \ell_s - h \geq 0$ the nonnegative overshoot, so $\mathbb{E}_{P_1}[T_h] \leq (h + \mathbb{E}[R_h])/I_1 = h/I_1 + O(1)$. Moreover, under the nominal law the nonnegative supermartingale $e^{W_t - t \cdot 0}$ built from the one-sided CUSUM and the change-of-measure $\mathbb{E}_\infty[e^{S_t}] = 1$ (with $S_t = \sum_{s \leq t} \ell_s$) give, via optional stopping at $T_c(h)$, the false-alarm floor $\text{ARL}_0(T_c(h)) \geq e^h = \gamma$.*

Proof. Finiteness of $\mathbb{E}_{P_1}[T_h]$: since $I_1 > 0$, the strong law gives $S_t/t \rightarrow I_1 > 0$ a.s. under P_1 , so $S_t \rightarrow +\infty$ and the level h is crossed in finite time a.s.; a standard renewal bound ($\mathbb{E}_{P_1}[T_h] \leq (h + c)/I_1$ for a constant c bounding the mean overshoot when ℓ_t has a finite second moment, which holds for the Gaussian surrogate) gives integrability. Wald's identity applies because T_h is a stopping time with $\mathbb{E}_{P_1}[T_h] < \infty$ and $\mathbb{E}_{P_1}|\ell_t| < \infty$: $\mathbb{E}_{P_1}[S_{T_h}] = I_1 \mathbb{E}_{P_1}[T_h]$. Writing $S_{T_h} = h + R_h$ with overshoot $R_h \geq 0$ and rearranging yields $\mathbb{E}_{P_1}[T_h] = (h + \mathbb{E}_{P_1}[R_h])/I_1$; the overshoot has bounded mean (again by finite second moment of ℓ_t), so $\mathbb{E}_{P_1}[T_h] = h/I_1 + O(1)$. For the false-alarm floor, note $M_t = e^{S_t}$ is a nonnegative P_0 -martingale because $\mathbb{E}_{P_0}[e^{\ell_t}] = \int (dP_1/dP_0) dP_0 = 1$. The one-sided statistic satisfies $W_t = \max_{0 \leq j \leq t} (S_t - S_j) \geq S_t - S_k$ for the last reset time k , and at the alarm $W_{T_c} \geq h$, so the post-reset walk has crossed h ; applying the optional-stopping inequality to M_t at the (bounded-truncation of the) crossing time and taking truncation to infinity gives $\mathbb{E}_\infty[T_c(h)] \geq e^h$. With $h = \log \gamma$ this is $\text{ARL}_0 \geq \gamma$. $\square$

C. Noncommutativity certifies detectability: a Fisher-information floor

We now prove the paper's central mechanism at the level of information, not just representation: the detectable drift signal is lower-bounded by the noncommutativity of channel selection with the causal shift, so the commutator that Theorem 1 identifies as the source of the kernel's expressivity is *the same* quantity that guarantees the change-point is statistically testable.

Parametrize the drift by a scalar amplitude $\theta \geq 0$: the drift law P_θ adds to the nominal feature covariance a rank-controlled lagged coupling supported at the informative lag τ^* ,

$$\Sigma_\theta = \Sigma_0 + \theta \Delta, \quad \Delta = \Delta_{\tau^*} \text{ supported on the lag-}\tau^* \text{ block,} \quad (6)$$

with $\theta = 0$ nominal and $\theta = 1$ the benchmark drift, so Δ is exactly the mean-shift Δ_{τ^*} of Theorem 2 lifted to the feature covariance. Define the *noncommutativity coefficient* of the drift as the norm of the part of Δ that no commutative rearrangement of channel and shift can remove,

$$\kappa := \|[S^{\tau^*}, E]\|_\Delta := \left(\sum_{c \neq c'} w_{\tau^*}^2 (\langle X_{:,c}, S^{\tau^*} X_{:,c'} \rangle - \langle X_{:,c'}, S^{\tau^*} X_{:,c} \rangle) \right) \quad (7)$$

where E denotes channel selection and $[S^{\tau^*}, E] = S^{\tau^*} E - E S^{\tau^*}$ is the commutator whose nonvanishing at $\tau^* \geq 1$ is the noncommutativity of Theorem 1; κ is the root-mean-square of the order-sensitive (antisymmetric-in- (c, c')) lagged correlations that a commutative monitor cannot see. At $n=1$, or whenever S^{τ^*} commutes with channel selection, $\kappa = 0$.

Theorem 8 (Noncommutativity lower-bounds the KL information and the Fisher information). *Let P_θ be the Gaussian CST-feature surrogate with covariance (6) and fixed mean, and let $\mathcal{I}(\theta) = -\mathbb{E}_\theta[\partial_\theta^2 \log p_\theta]$ be the Fisher information about the drift amplitude carried by one window. Then, with $\lambda_{\max} = \lambda_{\max}(\Sigma_0)$,*

$$\mathcal{I}(0) = \frac{1}{2} \text{tr}[(\Sigma_0^{-1} \Delta)^2] \geq \frac{\|\Delta\|_F^2}{2\lambda_{\max}^2} \geq \frac{\kappa^2}{2\lambda_{\max}^2}, \quad (8)$$

and the per-window KL information of the benchmark drift obeys $I_1 = D_{\text{KL}}(P_1 \| P_0) \geq \frac{1}{4} \mathcal{I}(0) (1 + o(1)) \geq \kappa^2 / (8\lambda_{\max}^2)$ for small amplitude. Consequently:

- (i) Nonzero noncommutativity certifies detectability. *If $\kappa > 0$ then $\mathcal{I}(0) > 0$ and $I_1 > 0$, so by Theorem 6 the CST-CUSUM detects the change point at finite worst-case delay $\text{ADD} \leq \log \gamma / I_1 \leq 8\lambda_{\max}^2 \log \gamma / \kappa^2$.*
- (ii) Vanishing noncommutativity makes the drift undetectable. *If $\kappa = 0$ (in particular for the commutative $n=1$ monitor), then the noncommutative coordinate is absent from the feature map, Δ projects to 0 in the monitor's coordinates, its $\mathcal{I}(0) = 0$ and $I_1 = 0$, the change point induces no measure change in the observed statistic, and no sequential test on that statistic attains a false-alarm floor $\gamma > 1$ at any finite delay — the detector is at chance, consistent with Theorem 2 and Proposition 4.*

Proof. For two zero-mean Gaussians with covariances Σ_θ and Σ_0 the KL divergence is $D_{\text{KL}}(P_\theta \| P_0) = \frac{1}{2} (\text{tr}(\Sigma_0^{-1} \Sigma_\theta) - d - \log \det(\Sigma_0^{-1} \Sigma_\theta))$. Substituting $\Sigma_\theta = \Sigma_0 + \theta \Delta$, put $A = \Sigma_0^{-1/2} \Delta \Sigma_0^{-1/2}$, so $\Sigma_0^{-1} \Sigma_\theta = I + \theta \Sigma_0^{-1} \Delta$

is similar to $I + \theta A$. Using $\text{tr}(I + \theta A) - d = \theta \text{tr} A$ and $\log \det(I + \theta A) = \text{tr} \log(I + \theta A) = \theta \text{tr} A - \frac{\theta^2}{2} \text{tr}(A^2) + O(\theta^3)$, the linear terms cancel and $D_{\text{KL}}(\hat{P}_\theta \| P_0) = \frac{\theta^2}{4} \text{tr}(A^2) + O(\theta^3)$. Since Fisher information equals twice the curvature of the KL divergence at the truth, $\mathcal{I}(0) = \partial_\theta^2 D_{\text{KL}}(P_\theta \| P_0)|_{\theta=0} = \frac{1}{2} \text{tr}(A^2) = \frac{1}{2} \text{tr}[(\Sigma_0^{-1} \Delta)^2]$, the first equality in (8). For the bound, A is symmetric and $\text{tr}(A^2) = \|A\|_F^2 = \|\Sigma_0^{-1/2} \Delta \Sigma_0^{-1/2}\|_F^2$. For symmetric positive definite B and any M one has $\|BMB\|_F \geq \sigma_{\min}(B)^2 \|M\|_F$, because $\text{vec}(BMB) = (B \otimes B) \text{vec}(M)$ and the smallest singular value of $B \otimes B$ is $\sigma_{\min}(B)^2$. Taking $B = \Sigma_0^{-1/2}$, so $\sigma_{\min}(B)^2 = \lambda_{\min}(\Sigma_0^{-1}) = \lambda_{\max}(\Sigma_0)^{-1}$, gives $\|A\|_F \geq \lambda_{\max}(\Sigma_0)^{-1} \|\Delta\|_F$, hence $\text{tr}(A^2) \geq \|\Delta\|_F^2 / \lambda_{\max}^2$ and $\mathcal{I}(0) \geq \|\Delta\|_F^2 / (2\lambda_{\max}^2)$. Finally $\|\Delta\|_F^2 \geq \kappa^2$ because κ^2 is the squared Frobenius norm of the antisymmetric-in- (c, c') part of the lag- τ^* block of Δ , a sub-block of Δ , so (8) follows. The KL/Fisher relation $I_1 = D_{\text{KL}}(P_1 \| P_0) = \frac{\theta^2}{4} \text{tr}(A^2) + O(\theta^3) = \frac{1}{4} \mathcal{I}(0) \theta^2 (1 + o(1))$ at $\theta = 1$ gives the stated floor for small amplitude.

Claim (i): $\kappa > 0 \Rightarrow \mathcal{I}(0) > 0 \Rightarrow I_1 > 0$ by (8), and Theorem 6 converts $I_1 > 0$ into the finite delay $\log \gamma / I_1 \leq 8\lambda_{\max}^2 \log \gamma / \kappa^2$. Claim (ii): if $\kappa = 0$, the lag- τ^* antisymmetric block of Δ vanishes; for the commutative monitor the feature map excludes every $\tau \geq 1$ coordinate, so the monitor's observed statistic ϕ_t has the same law under P_0 and P_1 (Theorem 2: equal marginals and equal instantaneous covariance). Then $\ell_t \equiv 0$ almost surely in the monitor's σ -field, $\mathcal{I}(0) = 0$, $I_1 = I'_1 = 0$, and the pre- and post-change measures on the observed stream coincide. By the data-processing inequality no statistic of that stream can have positive KL, so for any stopping time T adapted to it, $\mathbb{E}_\nu[(T - \nu + 1)^+]$ cannot be made finite while $\mathbb{E}_\infty[T] \geq \gamma > 1$: a test achieving $\text{ARL}_0 \geq \gamma$ must, by the same measure identity, have $\mathbb{E}_\nu[T \mid T \geq \nu] = \mathbb{E}_\infty[T \mid T \geq \nu]$, so its post-change behaviour is indistinguishable from pure false alarm and its detection probability in any fixed window equals its false-alarm probability — chance. $\square$

Corollary 9 (Cramér–Rao floor on drift-amplitude estimation). *Any unbiased estimator $\hat{\theta}_m$ of the drift amplitude from m independent windows has variance $\text{Var}(\hat{\theta}_m) \geq (m\mathcal{I}(0))^{-1} \geq 2\lambda_{\max}^2 / (m\kappa^2)$. Hence the estimable amplitude resolution improves as the noncommutativity κ grows and diverges as $\kappa \rightarrow 0$: a commutative monitor cannot estimate the drift amplitude at any finite rate, while the CST monitor with $\kappa > 0$ attains the parametric $1/\sqrt{m}$ rate with constant $\sqrt{2}\lambda_{\max}/\kappa$.*

Proof. Windows are independent, so the sample Fisher information is $m\mathcal{I}(0)$; the Cramér–Rao inequality gives $\text{Var}(\hat{\theta}_m) \geq (m\mathcal{I}(0))^{-1}$, and (8) bounds $\mathcal{I}(0) \geq \kappa^2 / (2\lambda_{\max}^2)$. Substituting yields the claim; $\kappa \rightarrow 0$ sends the floor to ∞ . $\square$

Remark 10. Theorem 8 upgrades Theorem 2 from a first-moment statement to an information-theoretic one: it is not merely that the commutative monitor's mean score

fails to separate, but that the pre- and post-change laws of its observed statistic are identical, so *no* test on it — linear or not, one-shot or sequential — can beat chance. Noncommutativity is therefore necessary in the strong, Neyman–Pearson sense, and the commutator norm κ of (7) is the exact currency in which detectability is denominated.

D. Forecast-risk bound under temporal dependence, and a calibrated interval

The forecasting head is an estimator with uncertainty, and its telemetry is dependent, not i.i.d.: Fig. 2b shows the detrended T_1 series is geometrically mixing (its autocorrelation decays like ϕ^h for the AR(1) factor of Theorem 5). We bound the finite-sample excess risk of the CST ridge forecaster over its population optimum under this dependence and attach a predictive interval with asymptotically correct coverage. Let the forecaster be the linear map $\hat{f}_\beta(\phi) = \beta^\top \phi$ on the CST feature ϕ of a window, fit by ridge with penalty $\alpha > 0$ on m overlapping training windows, and let $\beta^* = \arg \min_\beta R(\beta)$, $R(\beta) = \mathbb{E}[(y - \beta^\top \phi)^2]$, be the population optimum (the target whose single-lag specialization is Theorem 5). Assume the stream is stationary and β -mixing with coefficients $\beta_{\text{mix}}(k) \leq \bar{\beta} e^{-c_\beta k}$ (geometric mixing, implied by the AR(1) structure), and the features and target are bounded, $\|\phi\| \leq B_\phi$, $|y| \leq B_y$.

Theorem 11 (Excess forecast risk under geometric mixing). *Under the assumptions above, with probability at least $1 - \delta$ over the training stream the ridge forecaster $\hat{\beta}$ satisfies*

$$R(\hat{\beta}) - R(\beta^*) \leq \alpha \|\beta^*\|^2 + c (B_\phi^2 + B_\phi B_y) \sqrt{\frac{\log(1/\delta)}{m_{\text{eff}}}}, \quad m_{\text{eff}} = \frac{1}{1} \quad (9)$$

for a universal constant c , where m_{eff} is the mixing-deflated effective sample size. The bound is the i.i.d. rate $O(m^{-1/2})$ inflated by the dependence factor $\sqrt{m/m_{\text{eff}}}$, which is $O(1)$ whenever the mixing rate c_β is bounded away from 0; increasing the truncation order n enlarges the feature dimension but leaves the rate unchanged, matching the empirical flatness of forecast skill in n (Fig. 3c).

The proof (Appendix C2) blocks the dependent stream into near-independent segments (Bernstein for mixing sequences via Berbee coupling [62], Lemma 12) and adds the ridge bias term; the noncommutative coordinates enter only through B_ϕ and never through the rate, which is why forecasting is agnostic to n .

Lemma 12 (Bernstein concentration for geometrically mixing sums). *Let $\{Z_t\}$ be a stationary, zero-mean, β -mixing sequence with $|Z_t| \leq B$ and $\beta_{\text{mix}}(k) \leq \bar{\beta} e^{-c_\beta k}$.*

Then for the block size $q = \lceil (2/c_\beta) \log m \rceil$ there is a universal constant c_0 with

$$\Pr \left[\left| \frac{1}{m} \sum_{t=1}^m Z_t \right| \geq \varepsilon \right] \leq 2 \exp \left(-\frac{c_0 m \varepsilon^2}{q B^2} \right) + \frac{m}{q} \bar{\beta} e^{-c_\beta q}. \quad (10)$$

Choosing q as above makes the second term $O(m^{-1})$, so the sum concentrates at scale $\varepsilon = \Theta(B\sqrt{q \log(1/\delta)/m}) = \Theta(B\sqrt{\log(1/\delta)/m_{\text{eff}}})$.

Proof. Partition $\{1, \dots, m\}$ into $\lfloor m/q \rfloor$ consecutive blocks of length q and couple the odd (resp. even) blocks to an independent copy: by Berbee's coupling lemma the block sequence differs from an i.i.d. block sequence on an event of probability at most $(m/q) \beta_{\text{mix}}(q) \leq (m/q) \bar{\beta} e^{-c_\beta q}$, the second term of (10). On the complementary event the block sums are independent, each bounded by qB , so the classical Bernstein/Hoeffding inequality for the $\lfloor m/q \rfloor$ independent block sums gives the first term with the variance proxy qB^2 . Optimizing the block length as $q = \Theta(c_\beta^{-1} \log m)$ balances the two terms and yields the stated effective-sample-size scaling. $\square$

Corollary 13 (Calibrated predictive interval for the coherence forecast). *Let $\hat{q}_{1-\alpha}$ be the empirical $(1-\alpha)$ -quantile of the absolute forecast residuals $\{|y_i - \hat{\beta}^\top \phi_i|\}$ on a held-out calibration block of m_c windows drawn from the same stationary stream, and form the interval $\mathcal{C}(\phi) = [\hat{\beta}^\top \phi - \hat{q}_{1-\alpha}, \hat{\beta}^\top \phi + \hat{q}_{1-\alpha}]$. Then the coverage satisfies*

$$\Pr[y \in \mathcal{C}(\phi)] \geq 1 - \alpha - 2\beta_{\text{mix}}(1) - O(m_c^{-1/2}), \quad (11)$$

so the interval is asymptotically calibrated at level $1 - \alpha$, with a coverage gap controlled by the one-step mixing coefficient (which is exponentially small under geometric mixing) and the calibration-set size.

Proof. For exchangeable residuals the split-conformal guarantee gives exact coverage $\Pr[y \in \mathcal{C}(\phi)] \geq 1 - \alpha$ with the finite-sample correction $\lceil (1-\alpha)(m_c+1) \rceil / (m_c+1) = 1 - \alpha + O(m_c^{-1})$. The calibration residuals here are not exchangeable but stationary and β -mixing; coupling the calibration block to an i.i.d. block (Berbee, as in Lemma 12) changes the empirical residual quantile by at most the total-variation cost $2\beta_{\text{mix}}(1)$ per adjacent pair, and the Dvoretzky–Kiefer–Wolfowitz inequality controls the quantile estimation error of the coupled i.i.d. residual CDF at rate $O(m_c^{-1/2})$. Summing the two corrections gives (11). $\square$

E. A distribution-free anomaly-flagging rule with controlled false alarms

Finally we give the one-shot (per-window) counterpart of the sequential detector: a probabilistic anomaly rule that flags a bad-data / drift-excursion window while controlling the false-positive rate *without any distributional*

assumption on the nominal law — the data-quality guarantee a probabilistic surrogate must supply. The rule is a split-conformal test [63] that thresholds the CST reconstruction score $s(X) = \|(I - P_k)(\Phi_n(X) - \bar{\Phi})\|^2$ of Theorem 2 at a data-driven level.

Theorem 14 (Conformal false-positive control for CST anomaly flagging). *Fit the rank- k nominal subspace P_k on a training block, and let $s_{(1)} \leq \dots \leq s_{(m_c)}$ be the order statistics of the CST reconstruction scores on a disjoint calibration block of m_c nominal windows. For target level $\alpha \in (0, 1)$ set the threshold $h_\alpha = s_{(\lceil (1-\alpha)(m_c+1) \rceil)}$ and flag a test window X as anomalous iff $s(X) > h_\alpha$. If the calibration and test nominal windows are exchangeable, then the per-window false-positive rate is controlled distribution-free,*

$$\Pr_{P_0}[\text{flag}] = \Pr_{P_0}[s(X) > h_\alpha] \leq \alpha, \quad \text{and} \quad \Pr_{P_0}[s(X) > h_\alpha] \geq \alpha - 1/(m_c+1) \quad (12)$$

Under only stationarity and β -mixing the same bound holds up to an additive $2\beta_{\text{mix}}(1)$. Moreover the rule has nontrivial power against the drift: its detection probability is bounded below by $\Pr_{P_1}[s(X) > h_\alpha] \geq \alpha + (\text{energy of } (I - P_k)\Delta_{\tau^*}) / (\text{score dispersion}) - o(1)$, which is strictly positive whenever $\kappa > 0$ by Theorem 8, and equals α (chance) when $\kappa = 0$.

Proof. False-positive control is the standard split-conformal / rank argument. Under exchangeability of the $m_c + 1$ nominal scores $\{s_1, \dots, s_{m_c}, s(X)\}$, the rank of the test score $s(X)$ among them is uniform on $\{1, \dots, m_c + 1\}$. The event $\{s(X) > h_\alpha\}$ means $s(X)$ exceeds the $\lceil (1-\alpha)(m_c+1) \rceil$ -th order statistic, i.e. its rank is at least $\lceil (1-\alpha)(m_c+1) \rceil + 1$, an event of probability $(m_c + 1 - \lceil (1-\alpha)(m_c+1) \rceil) / (m_c + 1) \leq \alpha$; the matching lower bound $\geq \alpha - 1/(m_c + 1)$ follows from the ceiling rounding, giving (12). The distribution-free property is immediate: the argument uses only the rank, never the form of P_0 . Replacing exchangeability by stationary β -mixing costs one Berbee coupling of the test window to the calibration block, an additive $2\beta_{\text{mix}}(1)$ in total-variation, as in Corollary 13. For power, decompose $s(X) = \|(I - P_k)(\Phi_n(X) - \bar{\Phi})\|^2$; under P_1 the mean of the residual increases by $\|(I - P_k)\Delta_{\tau^*}\|^2$ (Theorem 2), so by Chebyshev the exceedance probability at the fixed nominal quantile h_α rises above α by the stated separation-to-dispersion ratio, which is positive iff the off-subspace drift energy is positive, i.e. iff $\kappa > 0$; when $\kappa = 0$ the residual law is unchanged and the exceedance probability stays at α . $\square$

Remark 15. The four theorems assemble into a single sequential-statistics pipeline for the CST surrogate: Theorem 14 screens each window for bad data at a controlled false-positive rate; Theorem 6 accumulates the surviving windows into a minimax-optimal change-point alarm; Theorem 8 certifies that the alarm fires at finite delay exactly when the drift carries noncommutativity $\kappa > 0$; and Theorem 11 with Corollary 13 turns the same features into a calibrated forecast. Statistics is thus not an

add-on but the structure of the monitor: change-point testing, Fisher information, and mixing concentration are the mechanisms by which a C^* -algebraic kernel becomes an operational reliability instrument.

IV. DISCUSSION

We have recast quantum-hardware reliability monitoring as operator-algebraic learning and distilled the result into a single operational rule: *match the kernel’s non-commutativity to the geometry of the drift*. Level-based and instantaneous monitors — including the periodic spectral-truncation kernel from which our construction descends — are blind to correlated decoherence that lives in ordered, lagged cross-channel structure, which is the very failure mode that quantum error correction is least able to absorb. The causal spectral-truncation kernel resolves it, and an elementary certificate (Theorem 3) both explains why a small truncation order is optimal and predicts, with no fitting, where that optimum sits. Because the kernel recovers established forecasting and low-rank detection at its commutative order, the family is a strict generalization of existing monitors rather than a competitor, and inherits their interpretability while extending their reach. Deep sequence-model monitors — recurrent [16, 17] and attention-based [18] architectures studied elsewhere for telemetry [14, 15] — are complementary but require training and many parameters, whereas the present kernel is training-free at inference, deterministic, and CPU-reproducible in seconds; the contribution is thus both a theoretical characterization of when noncommutative monitoring is necessary and a practical toolbox that runs on commodity hardware.

Several limitations bound these claims and define the natural next steps. The telemetry generator, although physically motivated, is a model rather than measured device data, and the correlated-decoherence mode is a controlled construction chosen to isolate the noncommutative effect in the cleanest possible setting; validation on archival calibration logs from operating processors and on a standard anomaly-detection benchmark [20], and a head-to-head comparison against the classical change-point detectors catalogued in [59] and the cumulative-sum chart [60], is the priority follow-up. The kernel as presented is linear in its feature map, so composing it with an outer nonlinearity, and pairing the forecaster with an explicit recalibration policy under a cost model and uncertainty-aware predictions [19], are concrete extensions toward a deployed monitor. Because the entire pipeline is open, hardware-agnostic, and fast to reproduce, it is positioned to serve as a common, auditable substrate for that work.

a. Outlook: monitoring as probabilistic surrogate modeling. Read broadly, the CST kernel is a step toward treating hardware monitoring as statistical, uncertainty-aware modeling of an expensive stochastic system. The present detector is unsupervised and

training-free, but its reconstruction score is a likelihood-ratio surrogate in disguise, and its natural successor is an explicit probabilistic surrogate — a Bayesian emulator of the calibration process that returns not a point score but a calibrated posterior over drift state, propagating the sampling noise inherent in every telemetry entry through to a recalibration decision. Such a surrogate would let the monitor integrate mechanistic knowledge of the noise channels it watches (the two-level-system and $1/f$ physics that shapes the drift) as a prior rather than relearning it, support active experimental design that spends characterization circuits only where the posterior is uncertain, and keep the interpretability that makes the noncommutative order a physically meaningful, auditable quantity rather than a tuned knob. The representation-complexity certificate we prove here is, in that light, a first instance of a broader principle: when the object being monitored is statistical to its core, the right amount of model structure is itself something to be estimated and certified, not assumed.

V. CONCLUSIONS

We have introduced the causal spectral-truncation kernel and shown, both empirically and through an elementary certificate, that the geometry of hardware drift dictates how much noncommutativity a reliability monitor must carry. A single truncation order interpolates from a commutative instantaneous monitor to a noncommutative one that reads ordered, lagged cross-channel structure: on smooth marginal drift the commutative member forecasts coherence with horizon-growing skill, whereas on correlated decoherence — the failure mode quantum error correction can least absorb — only the noncommutative member detects the event, at a truncation order the theory predicts with no fitting. The practical content is a single rule: match the kernel’s noncommutativity to the geometry of the drift.

Because the monitor is training-free, deterministic, and reproducible on one CPU in seconds, it can run as a passive layer over the calibration telemetry a processor already records, complementing rather than replacing the protocol-based characterization that diagnoses errors on demand [6–8]. As devices scale toward the qubit counts at which fault tolerance becomes feasible, reliability monitoring at that scale is itself a near-term bottleneck, and an operator-algebraic monitor whose cost grows only linearly in the noncommutativity it exploits — and whose optimal complexity is certified rather than tuned — is a concrete step toward meeting it. We expect the same representation-complexity trade-off to organize the monitoring of other correlated-error channels, and the kernel to compose naturally with the nonlinear and uncertainty-aware extensions outlined above.

VI. METHODS

a. Telemetry generation. We generate a synthetic multi-qubit dataset for $Q = 5$ qubits over $N = 200$ half-hour steps. For qubit q at time t (in hours) the relaxation time is $T_1(q, t) = \max(10, 80 + 20 \sin(2\pi t/48 + 0.8q) - 0.05t + \varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, 2^2)$, superposing a slow qubit-dependent oscillation, a secular decay, and Gaussian fluctuation. The dephasing time is $T_2 = 0.9T_1\eta$ with $\eta \sim \mathcal{N}(1, 0.04^2)$, and the single- and two-qubit gate fidelities, readout-assignment error, error-per-Clifford, and cross-resonance phase are generated by analogous bounded processes carrying small secular trends, so that all seven channels track T_1 by construction. A binary drift label marks any step with $T_1 < 50\mu\text{s}$ or single-qubit fidelity below 0.998. Generation is fully seeded and therefore bit-for-bit reproducible.

b. Correlated-decoherence benchmark. The detection benchmark contrasts nominal windows with *correlated-decoherence* windows that share the same physical base trajectory B and the same per-channel noise scales σ . The drift class injects a single shared *white* latent u into four channels (T_1 , T_2 , readout error, cross-resonance phase) at channel-specific integer delays $(0, 1, 2, 3)$ under variance-preserving mixing $x_c \leftarrow B_c + \sigma_c(\sqrt{1 - a^2}\varepsilon_c + a u_{t-\delta_c})$ with $a = 1$. Because the latent is white, every per-channel marginal variance and every temporal autocorrelation are left unchanged, and the population lag-0 cross-covariance ($\tau = 0$, $\delta_c \neq \delta_{c'}$) is zero in both classes; the entire perturbation is carried by the ordered lag- τ cross-channel covariance ($\tau = \delta_{c'} - \delta_c \geq 1$). Per-window temporal centering over $L = 32$ leaves a small finite-window residual at $\tau = 0$, but the drift energy is concentrated at $\tau \geq 1$ (Fig. 2c). The benchmark is therefore the cleanest possible separator: the commutative monitors we evaluate — raw-level, instantaneous-covariance, and the periodic spectral-truncation kernel — are all at chance and the drift is exposed only by the causal noncommutative ($n \geq 2$) kernel, which is the empirical content of Theorem 2.

c. Causal spectral-truncation kernel. A window $X \in \mathbb{R}^{L \times C}$ ($L = 32$, $C = 7$) is centered in time per channel and mapped to the feature tensor $\Phi_n(X)_{cc'\tau} = \langle X_{:,c}, S^\tau X_{:,c'} \rangle / (L - \tau)$ for $\tau = 0, \dots, n-1$, with S the causal lower shift; the operator-valued kernel is (1) and the scalar kernel is the Hilbert–Schmidt inner product of these features. The commutative member ($n=1$) retains only the instantaneous covariance, and each $n>1$ adds one band of ordered lagged cross-channel coordinates. For comparison we implement the *periodic* spectral-truncation baseline unchanged: it maps each channel to the $n \times n$ Toeplitz matrix of its truncated Fourier coefficients and forms the ordered cross-channel products of those matrices, exactly the construction of Hashimoto et al. [9]. The two kernels differ only in whether the truncated algebra is the causal shift (ours) or the periodic Fourier multiplier (theirs), which isolates causality as the operative ingredient.

d. Forecasting. We compare persistence (carry the last T_1 forward), climatology (the training-mean horizon trajectory), an autoregressive ridge on the 32-step T_1 history, and a multivariate ridge [65] on the flattened 32×7 window (ridge penalty $\alpha = 5$; this is the commutative $n=1$ kernel member), under a leak-free per-qubit 80/20 chronological split, scoring MAE per horizon step over five generator seeds with skill $1 - \text{MAE}/\text{MAE}_{\text{persistence}}$. The CST forecaster augments the window with the order- n causal features before the ridge solve; its eight-step skill versus n (Fig. 3c) is averaged over the same five seeds. The chronological split guarantees that no future sample informs a past prediction.

e. Detection. The unsupervised detector standardizes the kernel feature map on nominal training windows, fits a rank- k principal subspace by principal component analysis [66], and scores each test window by its mean squared reconstruction residual outside that subspace. Detection quality is ROC-AUC against the drift labels over eight randomized 70/30 splits; because the detector is fit on nominal windows only, randomization cannot leak a label it never uses. The truncation sweep reports, per order n , the best ROC-AUC over bottleneck ranks $k \in \{1, 2, 3, 5, 8, 12, 16\}$, and the kernel comparison (Table II) fixes $k = 8$ so that all detectors are compared at matched representational budget.

f. Computational complexity and resource accounting. Each window $X \in \mathbb{R}^{L \times C}$ is mapped to a feature tensor $\Phi_n(X)$ of nC^2 coordinates, every coordinate an inner product of length at most L , so feature extraction costs $O(nC^2L)$ time and $O(nC^2)$ memory per window — linear in the window length and the truncation order, and quadratic only in the channel count C (here $C=7$, $L=32$, and $n \leq 16$, for at most 784 feature coordinates). Fitting the detector standardizes these features and extracts a rank- k principal subspace of the $N_{\text{train}} \times nC^2$ nominal feature matrix, an $O(N_{\text{train}} nC^2 \min(N_{\text{train}}, nC^2))$ singular-value decomposition; scoring a test window then costs $O(nC^2L)$ to featurize and $O(nC^2k)$ to project onto the subspace and measure the residual. Every step is a dense linear-algebra primitive with no iterative optimization, no gradients, and no learned parameters, so the entire truncation-and-rank sweep is polynomial and runs in seconds on a single CPU core. This is the resource counterpart of the operating-characteristic result: the monitor is training-free at inference, and its cost grows only linearly in the noncommutativity order n that it is permitted to exploit.

g. Reproducibility and compute. The analyses run entirely on a single CPU and regenerate in seconds, with no GPU and no training. Every figure and table is computed live from the installed `qdriftforecast` package by `make_paper_figures.py` (the installed `qdrift-reproduce` entry point), and reported uncertainties are mean $\pm$ standard deviation across the $n = 5$ generator seeds (forecasting) or $n = 8$ randomized splits (detection). Determinism is byte-level: the seeds and the build epoch are pinned and the vector

PDF output is fixed across runs, so every number in the manuscript is a reproducible function of integer seeds.

CODE AVAILABILITY

The telemetry generator, the causal spectral-truncation kernel, the detection and forecasting modules, and the figure-and-table pipeline are openly available in the project repository under an MIT license as the installable `qdriftforecast` package. All figures, tables, and reported uncertainties regenerate from the released code on commodity CPU hardware in seconds.

Appendix A: Extended data tables

Appendix B: Supplementary derivations and reproduction details

This appendix is a self-contained derivation of every concept the released code implements, written for a reader with linear algebra and basic probability but no prior exposure to quantum-hardware telemetry, operator-algebraic kernels, or the estimators used here. Every symbol that appears in the code is defined, and each subsection names the source file (under `submission/code/`) that realizes it, so the appendix doubles as a guide to the artifact. The whole pipeline is CPU-only and deterministic: every number in the paper is a fixed function of integer seeds.

a. Source-file map. All paths are relative to `submission/code/`: `qdriftforecast/data.py` (telemetry generator, correlated-drift benchmark, windowing, splitting); `qdriftforecast/kernels.py` (the causal spectral-truncation kernel, its operator- and scalar-valued forms, and the periodic baseline); `qdriftforecast/detection.py` (the reconstruction detector, truncation sweep, (n, k) grid and kernel comparison); `qdriftforecast/forecasting.py` (forecasters); `make_paper_figures.py` (all figures and table bodies); and `generated_data/*.json` (the numbers behind each panel).

S1. The problem in one paragraph

A quantum processor is not static: its coherence times, gate fidelities, readout error, and coupling parameters drift over hours, eroding the error budget that quantum error correction depends on, so an operator must decide *when* a device has drifted and *which* intervals warrant recalibration. We pose two tasks on the multivariate telemetry — *forecast* near-future coherence, and *detect* drifted windows without labels — and the central finding concerns the second: one geometry of drift, correlated decoherence living in ordered, lagged cross-channel struc-

ture, leaves every commutative monitor we evaluate at chance and is detectable only by a noncommutative one.

S2. Computation model: telemetry as arrays

(*qdriftforecast/data.py.*) A telemetry record is a finite table in which each row is a `(timestamp, qubit_id)` pair holding measured or simulated features. The seven channels (`FEATURE_COLS`) are `T1_us`, `T2_us`, `gate_fidelity_1q`, `gate_fidelity_2q`, `readout_error`, `gate_error_per_clifford`, and `cross_resonance_phase_rad`. Every result in the paper is a deterministic function of these arrays and fixed integer seeds.

S3. The physics of the generator

(*generate_synthetic_dataset.*) For qubit q at time t the relaxation time is $T_1(q, t) = \max(10, 80 + 20 \sin(2\pi t/48 + 0.8q) - 0.05t + \varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, 2^2)$, superposing a 48-hour oscillation (recalibration-like cycling, with phase $0.8q$ decorrelating the qubits), a secular decay (aging), and Gaussian fluctuation. The remaining channels track T_1 by construction, and a binary drift label marks any step with $T_1 < 50 \mu\text{s}$ or single-qubit fidelity below 0.998. Because the deterministic part is smooth and the noise small, the detrended series is weakly stationary with a slowly decaying autocorrelation — the structure that S8 converts into forecast skill.

S4. The correlated-decoherence benchmark, and why commutative monitors are at chance

(*generate_correlated_drift_windows.*) Both classes share the same base trajectory B and noise scales σ . The drift class adds a shared white latent u to four channels at delays $\delta_c \in \{0, 1, 2, 3\}$ under variance-preserving mixing $x_c = B_c + \sigma_c(\sqrt{1 - a^2}\varepsilon_c + a u_{t-\delta_c})$. Because u is white, three matchings hold exactly: (i) each channel’s marginal variance is σ_c^2 in both classes; (ii) each channel’s temporal autocorrelation is unchanged, since a sum of independent white processes is white; and (iii) the *population* lag-0 cross-covariance between two injected channels is $a^2\sigma_c\sigma_{c'}\mathbb{E}[u_{t-\delta_c}u_{t-\delta_{c'}}] = 0$ for $\delta_c \neq \delta_{c'}$, matching nominal. The signal energy resides entirely in the lag- τ cross-covariance with $\tau = \delta_{c'} - \delta_c \geq 1$. Two caveats bound the scope of this matching: per-window temporal centering over the finite $L = 32$ window leaves a small $O(1/L)$ residual in the centered lag-0 cross-covariance of the drift class, so the matching is exact only in the stationary limit; and Theorem 2 is a statement about *first-moment* discriminability, not about every conceivable nonlinear statistic. The operative claim is the one we verify empirically: the commutative monitors we evaluate —

TABLE III. **Detection ROC-AUC versus truncation order.** Mean $\pm$ s.d. over the $n = 8$ randomized splits; best ROC-AUC over bottleneck ranks $k \in \{1, \dots, 16\}$ per order. Detection is at chance at the commutative order $n=1$, peaks sharply at $n=2$ (in **bold**), and decays monotonically toward chance as the representation becomes over-complete — the certified mechanism of Theorem 3 and the source data for Fig. 4a.

Order n	ROC-AUC	Order n	ROC-AUC
1	0.513 ± 0.041	6	0.692 ± 0.024
2	0.903 ± 0.016	8	0.647 ± 0.026
3	0.832 ± 0.022	10	0.619 ± 0.028
4	0.776 ± 0.023	12	0.596 ± 0.026
5	0.728 ± 0.023	16	0.573 ± 0.025

TABLE IV. **Per-horizon forecasting error.** Mean absolute error in μs , mean $\pm$ s.d. over the $n = 5$ telemetry-generator seeds. Each row is one step of the eight-step T_1 horizon, and the best (lowest) value per row is in **bold**. The learned ridge forecasters stay nearly flat across the horizon while persistence degrades monotonically and climatology is uniformly poor — the per-step view underlying Fig. 3a and the source of the widening skill margin.

Horizon (steps)	Persistence	Climatology	AR-ridge	Multivariate ridge
1	2.44 ± 0.16	13.01 ± 0.05	1.85 ± 0.07	1.80 ± 0.09
2	2.70 ± 0.11	12.89 ± 0.05	1.88 ± 0.08	1.82 ± 0.12
3	3.21 ± 0.08	12.79 ± 0.07	1.91 ± 0.05	1.84 ± 0.11
4	3.78 ± 0.17	12.69 ± 0.06	1.96 ± 0.07	1.84 ± 0.11
5	4.47 ± 0.11	12.63 ± 0.06	2.04 ± 0.06	1.88 ± 0.10
6	5.25 ± 0.19	12.55 ± 0.06	2.11 ± 0.05	1.91 ± 0.10
7	6.05 ± 0.14	12.49 ± 0.04	2.14 ± 0.07	1.90 ± 0.08
8	6.79 ± 0.18	12.40 ± 0.05	2.19 ± 0.06	1.88 ± 0.06

the raw-level detector, the commutative ($n=1$) kernel, and the periodic spectral-truncation kernel — are all at chance ($n=1$ ROC-AUC 0.51), because the drift energy is concentrated in the ordered, lagged cross-channel correlations they cannot represent (Fig. 2c).

S5. The causal spectral-truncation kernel, from scratch

(*qdriftforecast/kernels.py*.) Center each channel of a window in time. For lags $\tau = 0, \dots, n-1$ form the ordered cross-channel correlation $\Phi_n(X)_{cc'\tau} = \frac{1}{L-\tau} \sum_{t \geq \tau} X_{t,c} X_{t-\tau,c'}$, which are the matrix elements of $X^\top S^\tau X$ with S the causal lower shift. The operator-valued kernel $K_n(X, Y) = \sum_{\tau < n} w_\tau X^\top S^\tau Y \in M_C$ is an element of the channel C^* -algebra, and the scalar kernel is $\langle \Phi_n(X), \Phi_n(Y) \rangle$. At $n=1$ this is the symmetric instantaneous covariance and is commutative; for $n \geq 2$ the lagged terms are asymmetric in (c, c') because $S^\top \neq S$, which is exactly the noncommutativity the result turns on. Truncating the shift algebra to n powers is the causal analog of truncating the Fourier-multiplier algebra to n modes in the spectral-truncation construction of Hashimoto et al., and the periodic baseline (*periodic_truncation_feature_map*) implements that original torus construction unchanged, so the comparison in Fig. 4c varies only causality.

S6. The detector and the optimal-truncation certificate

(*qdriftforecast/detection.py*.) Standardize the kernel features on nominal windows, fit a rank- k principal subspace, and score a test window by its residual energy outside it. Increasing the truncation order n simultaneously captures more drift signal (good) and admits C^2 noise coordinates per lag (bad), so the separation-to-noise ratio is unimodal in n , peaking just past the informative lag (Theorem 3). Empirically the peak is at $n=2$ with ROC-AUC 0.90; the commutative endpoint and the over-complete limit are both at chance, and the rank axis of the same certificate recovers the classical low-rank over-completeness collapse.

S7. Metrics: MAE for forecasting, ROC-AUC for detection

Forecasting uses mean absolute error per horizon step, reported step by step so the widening gap over persistence is visible, with skill $1 - \text{MAE}/\text{MAE}_{\text{pers}}$. Detection uses ROC-AUC, which by the Mann–Whitney identity (Proposition 4) equals the probability that a random drift window outranks a random nominal one, is invariant under per-feature rescaling, and equals $\frac{1}{2}$ for a constant score — the precise chance level that the commutative and over-complete monitors attain.

S8. Why forecast skill grows with the horizon

This is Theorem 5. For a weakly stationary process with autocorrelation $\rho(h)$, the persistence risk is $2\gamma_0(1 - \rho(h))$ and the best single-lag predictor's risk is $\gamma_0(1 - \rho(h)^2)$, so the skill is $(1 - \rho(h))/2$, which increases as ρ decays. The multivariate ridge uses the full history and exceeds this single-lag bound, reaching 72% at horizon eight, while the noncommutative coordinates add nothing because the marginal structure is already linear and instantaneous (S5).

S9. How to reproduce everything

Install with `pip install qdriftforecast` (or, from a checkout, `pip install submission/code`). Run `qdrift-reproduce` (equivalently `python -m qdriftforecast.reproduce` or `python make_paper_figures.py` from `submission/code/`). The script writes, deterministically from fixed seeds on a single CPU: the five vector PDF figures into `./figures/`; the three table bodies into `./tables/` (`forecasting_benchmark.tex`, `detection_benchmark.tex`, `truncation_sweep.tex`); and the machine-readable source data into `generated_data/*.json`. No number in the manuscript is typed by hand; each is emitted by this pipeline.

S10. What the result means, and what it does not

A small, training-free, noncommutative kernel detects a correlated-decoherence drift that every commutative monitor — including the periodic spectral-truncation kernel — misses entirely, and a single truncation order certifies how much noncommutativity to use. Coherence forecasting succeeds wherever autocorrelation persists, and there noncommutativity is unnecessary. We make no quantum-hardware superiority claim: the telemetry is a physically motivated model and the correlated-decoherence mode is a controlled construction chosen to isolate the noncommutative effect. What the work delivers is a reusable, leakage-controlled, CPU-reproducible, operator-algebraic foundation for quantum-processor reliability monitoring, together with a demonstration — both empirical and certified — that the geometry of the drift dictates the noncommutativity a monitor must have.

Appendix C: Proofs for the sequential-detection theory

This appendix collects the two longer proofs deferred from Sec. III: the Lorden-minimax optimality and delay bound of the CST-CUSUM detector (Theorem 6), and the mixing-robust forecast-risk bound (Theorem 11). Both are self-contained given Lemmas 7 and 12.

1. Proof of Theorem 6 (CST-CUSUM minimax optimality)

Proof. The claim has two parts: an upper bound on the CST-CUSUM delay and a matching information lower bound over all admissible tests, both at threshold $h = \log \gamma$.

False-alarm floor. By Lemma 7, the exponential-martingale change-of-measure gives $\text{ARL}_0(T_c(h)) \geq e^h = \gamma$, so $T_c(\log \gamma)$ is admissible for the constraint $\text{ARL}_0 \geq \gamma$.

Delay upper bound. Because the CUSUM resets at 0 and the worst change time places the alarm just after the last reset, Lorden's reduction bounds the worst-case delay (2) by the mean first passage of the post-change random walk above h started at 0: $\text{ADD}(T_c(h)) \leq \mathbb{E}_{P_1}[T_h]$ with $T_h = \inf\{t : \sum_{s \leq t} \ell_s \geq h\}$. Lemma 7 evaluates this by Wald's identity as $\mathbb{E}_{P_1}[T_h] = (h + \mathbb{E}_{P_1}[R_h])/I_1 = h/I_1 + O(1) = \log \gamma/I_1 + O(1)$, since the overshoot R_h has bounded mean for the Gaussian surrogate increments. Dividing by $\log \gamma$, the $O(1)$ term is $o(\log \gamma/I_1)$, giving the upper half of (5).

Delay lower bound (Lorden). Lorden's theorem states that for any stopping time T with $\text{ARL}_0(T) \geq \gamma$, $\text{ADD}(T) \geq (\log \gamma)/I_1 (1 + o(1))$ as $\gamma \rightarrow \infty$, whenever the per-window KL information I_1 is finite and positive. We reproduce the argument. Fix the change point at $\nu = 1$ and let T be admissible. A change of measure from $P_0^\otimes$ to $P_1^\otimes$ on the event $\{T \leq t\}$ costs the likelihood ratio e^{S_t} , $S_t = \sum_{s \leq t} \ell_s$; by Wald's likelihood-ratio identity and Jensen,

$$\mathbb{E}_1[S_{T \wedge t}] \leq \log \frac{1}{\Pr_\infty[T \leq t]} (1 + o(1)) \leq \log \gamma (1 + o(1)),$$

where the last step uses $\Pr_\infty[T \leq t] \leq t/\text{ARL}_0 \leq t/\gamma$ over the relevant horizon $t = O(\text{ADD})$ and Markov's inequality on T . On the other hand $\mathbb{E}_1[S_T] = I_1 \mathbb{E}_1[T]$ by Wald's identity again, so $I_1 \mathbb{E}_1[T] \geq \log \gamma (1 + o(1))$, i.e. $\mathbb{E}_1[T] \geq \log \gamma/I_1 (1 + o(1))$. Taking the supremum over change times (which only increases the delay) gives $\text{ADD}(T) \geq \log \gamma/I_1 (1 + o(1))$, the lower half of (5).

Conclusion. The upper and lower bounds coincide to first order at $h = \log \gamma$, so $T_c(\log \gamma)$ attains the minimax delay and $\text{ADD}^*(\gamma) \sim \log \gamma/I_1$. This is exactly the delay-false-alarm tradeoff (5), with the CST feature map entering only through I_1 , which Theorem 8 lower-bounds by the noncommutativity κ . $\square$

2. Proof of Theorem 11 (excess risk under mixing)

Proof. Write the ridge objective $R_\alpha(\beta) = R(\beta) + \alpha \|\beta\|^2$ and its empirical version $\hat{R}_\alpha(\beta) = \hat{R}(\beta) + \alpha \|\beta\|^2$ with $\hat{R}(\beta) = \frac{1}{m} \sum_i (y_i - \beta^\top \phi_i)^2$. Let $\hat{\beta} = \arg \min \hat{R}_\alpha$ and $\beta^* = \arg \min R$. We control the excess risk by a uniform deviation of $\hat{R}$ from R plus the ridge bias. For any β in the ridge-feasible ball $\mathcal{B} = \{\|\beta\| \leq \|\beta^*\|\}$ (the ridge solution lies in $\mathcal{B}$ because $\hat{R}_\alpha(\hat{\beta}) \leq \hat{R}_\alpha(0) = \hat{R}(0)$ bounds

$\alpha\|\hat{\beta}\|^2 \leq \hat{R}(0)$ and, by the same inequality in population after the deviation step, keeps $\|\hat{\beta}\|$ within a constant factor of $\|\beta^*\|$; we absorb that constant into c), the loss $Z_t(\beta) = (y_t - \beta^\top \phi_t)^2$ is bounded by $B := (B_y + B_\phi \|\beta^*\|)^2$. Apply Lemma 12 to the centred process $Z_t(\beta) - \mathbb{E}Z_t(\beta)$ at each β and take a union bound over a $1/m$ -net of $\mathcal{B}$ (metric entropy $O(d \log m)$, $d = nC^2$): with probability $\geq 1 - \delta$,

$$\sup_{\beta \in \mathcal{B}} |\hat{R}(\beta) - R(\beta)| \leq cB \sqrt{\frac{\log(1/\delta) + d \log m}{m_{\text{eff}}}}, \quad m_{\text{eff}} = \frac{1}{1 + 2 \sum_{k \geq 1} \beta_{\text{mix}}(k)}.$$

Under geometric mixing $\sum_{k \geq 1} \beta_{\text{mix}}(k) \leq \bar{\beta} \sum_{k \geq 1} e^{-c_\beta k} = \bar{\beta}/(e^{c_\beta} - 1) < \infty$, so $m_{\text{eff}} = \Theta(m(1 - e^{-c_\beta}))$ as claimed. Combining the uniform deviation with the standard ridge inequality $R(\hat{\beta}) - R(\beta^*) \leq \alpha\|\beta^*\|^2 + 2 \sup_{\beta \in \mathcal{B}} |\hat{R}(\beta) - R(\beta)|$ (the factor 2 from evaluating the deviation at both $\hat{\beta}$ and β^*) gives (9) after absorbing the $\sqrt{d \log m}$ metric-entropy factor into the constant c and the leading $\log(1/\delta)$ dependence; the feature dimension $d = nC^2$ enters only inside the logarithm, so the $O(m_{\text{eff}}^{-1/2})$ rate is unchanged by the truncation order n . This is the analytic counterpart of the empirical flatness of forecast skill in n (Fig. 3c): the noncommutative coordinates enlarge d only logarithmically and never the rate. $\square$

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